

Electronics & Communication Systems

ZyBo Development Board
Xilinx VIVADO



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Agenda

- 1) ZyBo Development Board
- 2) Zynq 7000 APSoC
- 3) DDFS Implementation on ZyBo
- 4) Xilinx VIVADO Design Suite
- 5) Vivado Design Flow

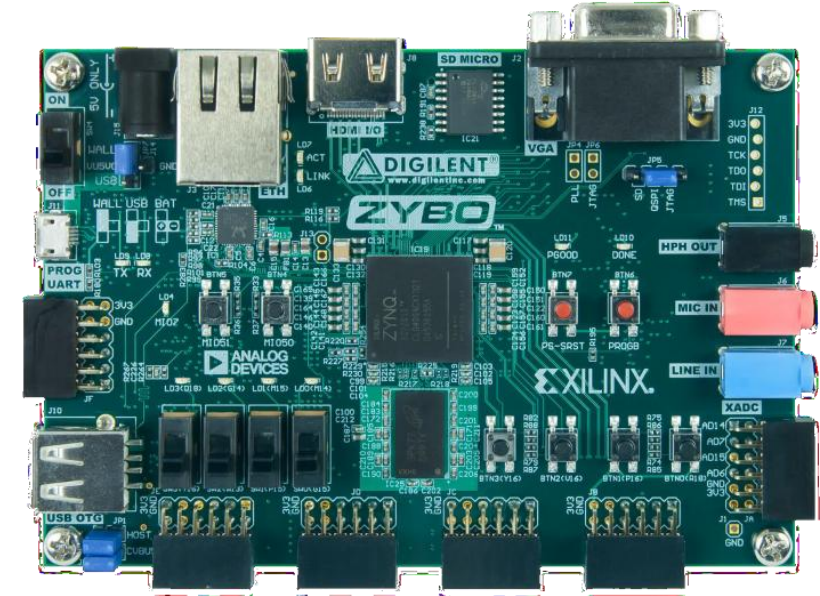
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ZyBo Development Board

The ZYBO (ZYNq BOard) is a feature-rich, ready-to-use, entry-level embedded software and digital circuit development platform built around the smallest member of the Xilinx Zynq-7000 family, the Z-7010.

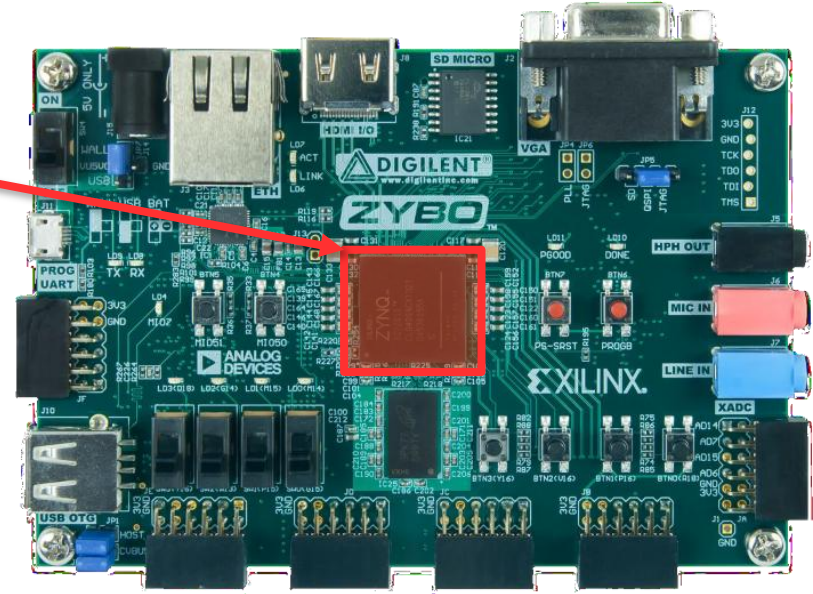
The Z-7010 is based on the Xilinx All Programmable System-on-Chip (AP SoC) architecture, which tightly integrates a dual-core ARM Cortex-A9 processor with Xilinx 7-series Field Programmable Gate Array (FPGA) logic.



ZyBo Development Board

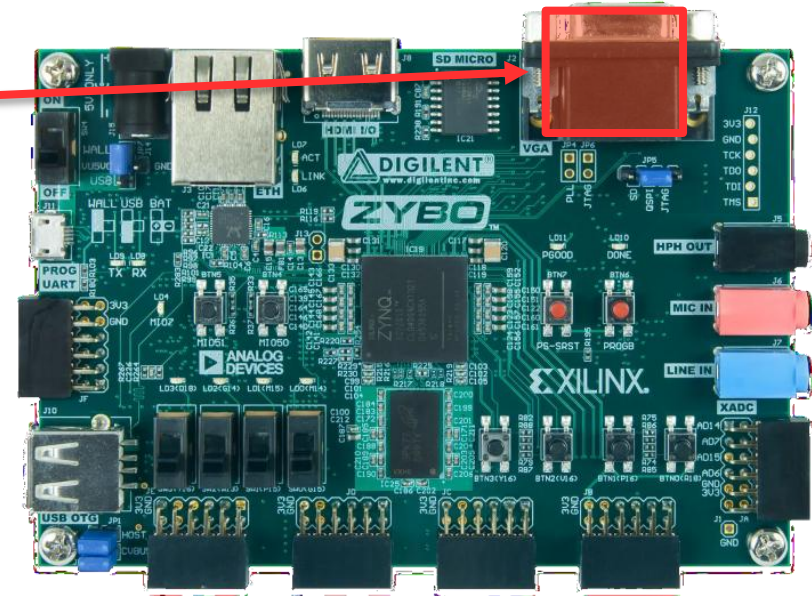
- ZYNQ XC7Z010-1CLG400C
- Others: ZyBo datasheet

https://www.xilinx.com/support/documentation/university/XUP%20Boards/XUPZYBO/documentation/ZYBO_RM_B_V6.pdf



ZyBo Development Board

- ZYNQ XC7Z010-1CLG400C
- 16-bits VGA port



ZyBo Development Board

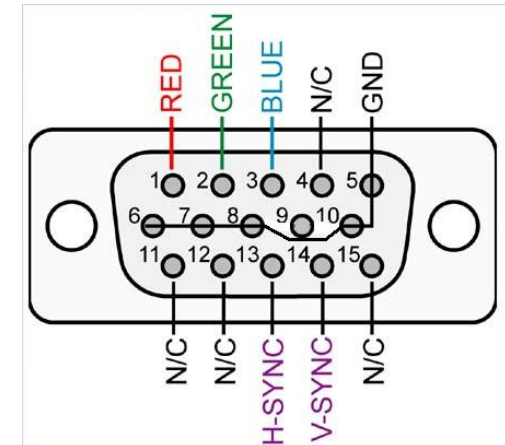
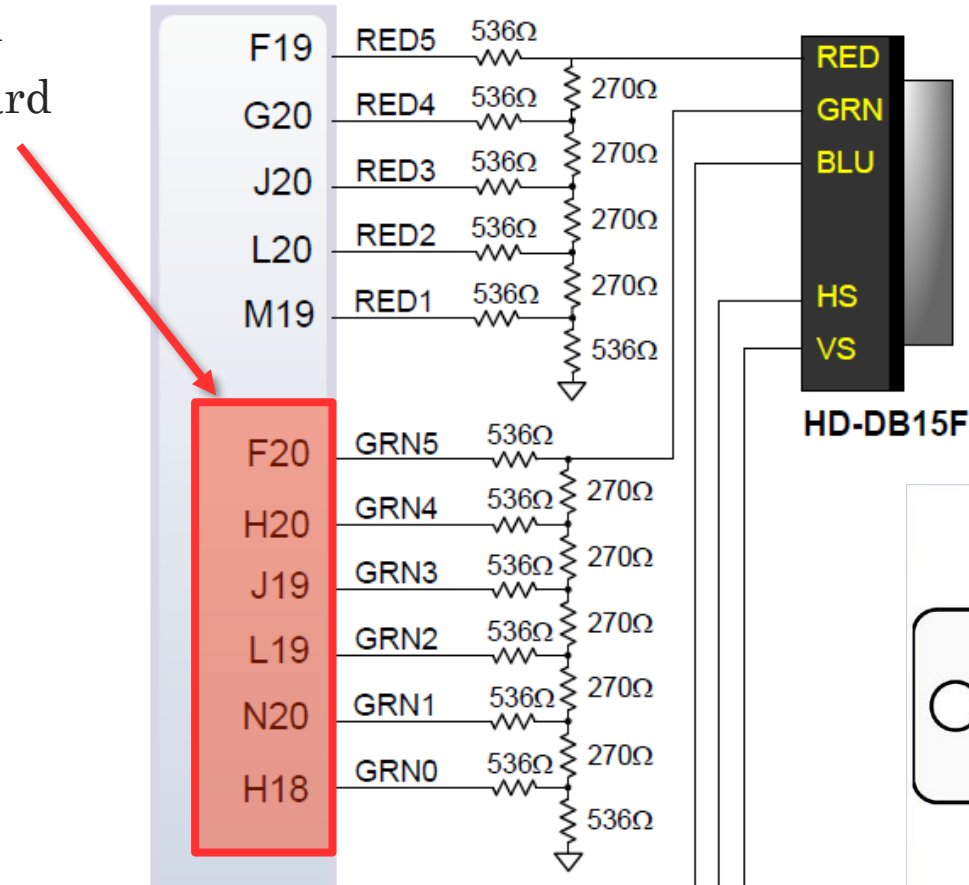
The VGA Port:

- 6 bits DAC on GRN channel
- R-2R ladder soldered on board

ZYNQ I/O pin VIO = 3.3 V

If $\text{GRN}[5:0] = \text{"111111"}$

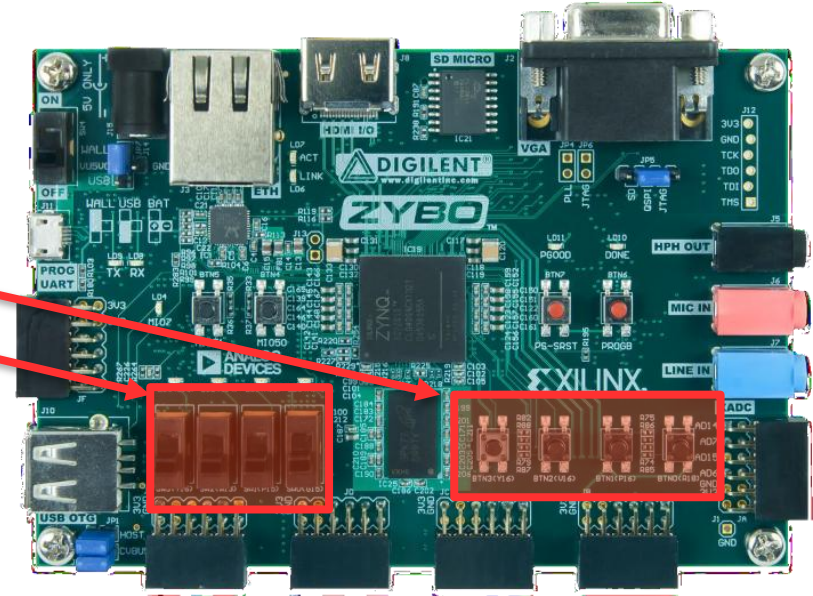
$V_{DAC}(63) \approx 3.2 \text{ V}$



Bottom view!

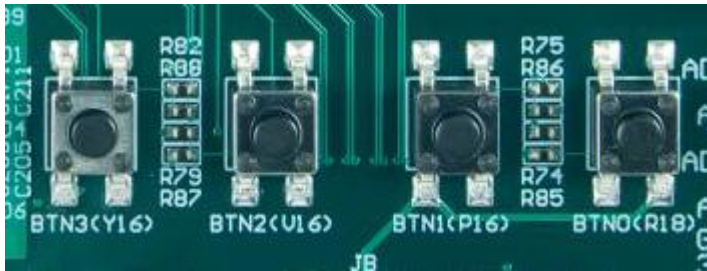
ZyBo Development Board

- ZYNQ XC7Z010-1CLG400C
- 16-bits VGA port
- GPIO:
 - 4 pushbuttons
 - 4 slide switches
 - 4 LEDs

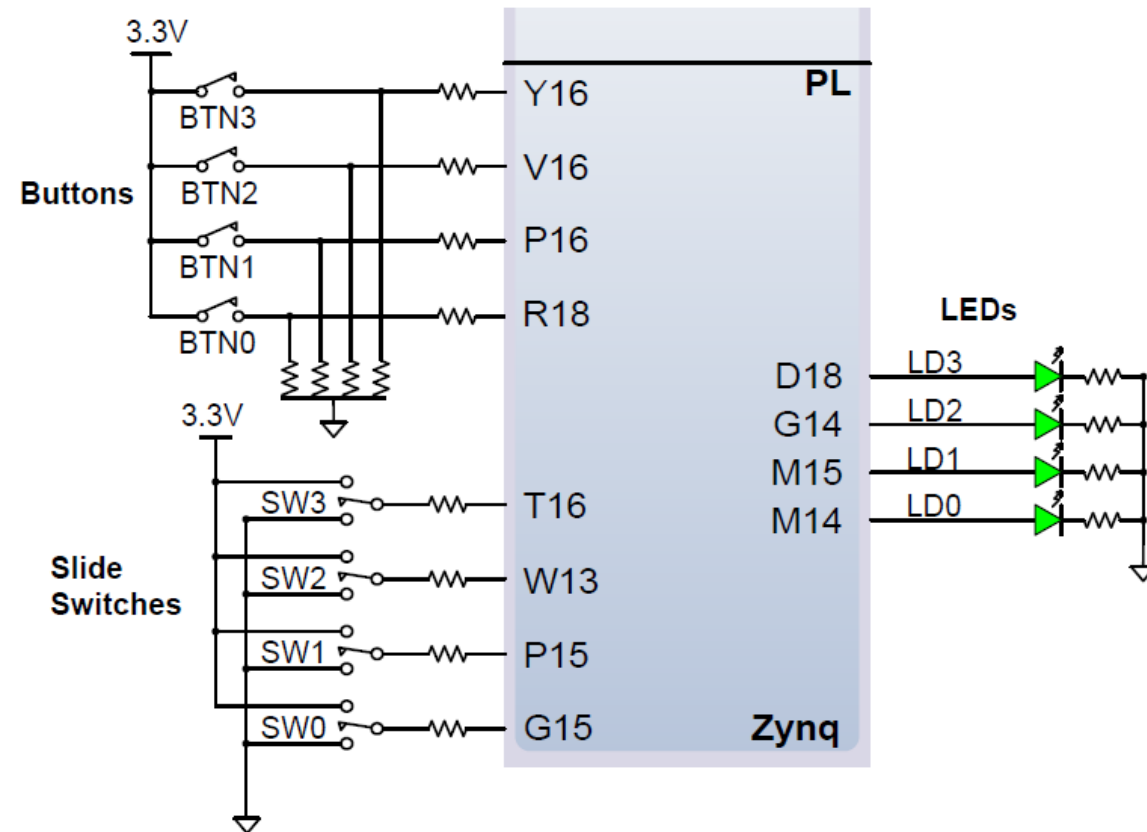
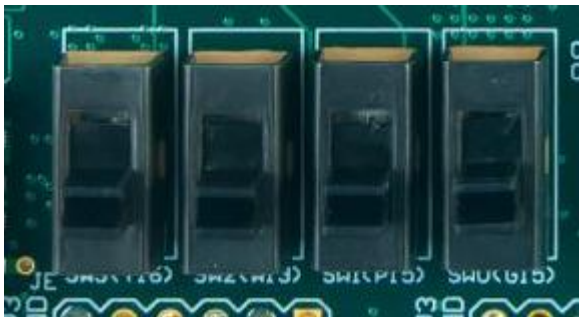


ZyBo Development Board

Push Buttons

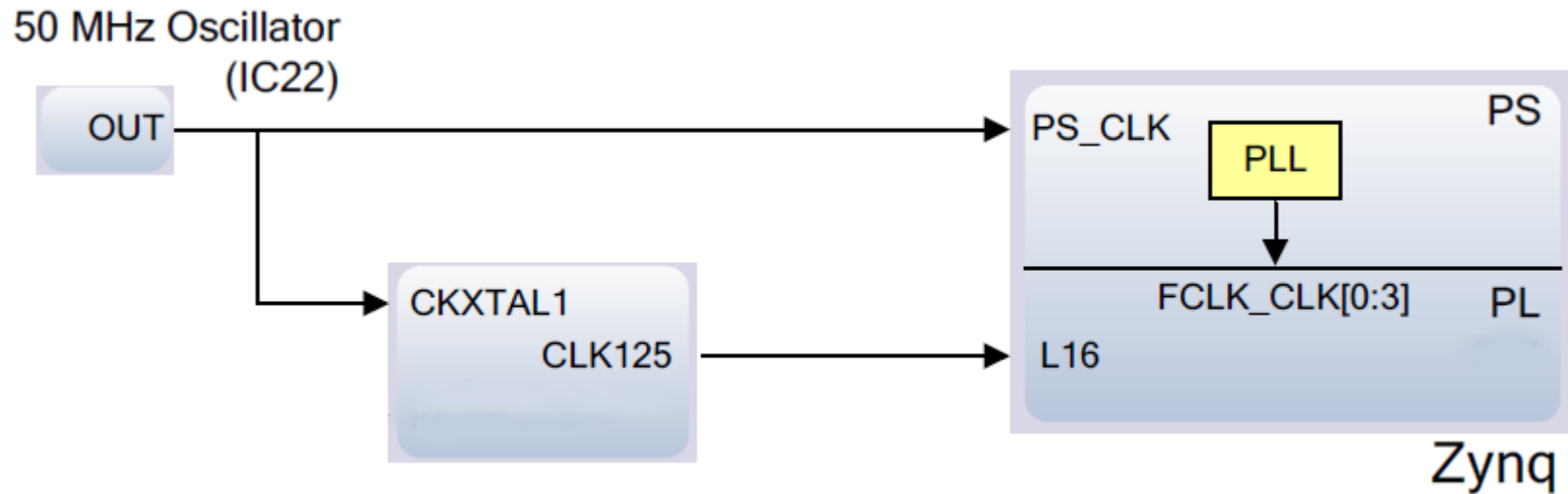


Switches



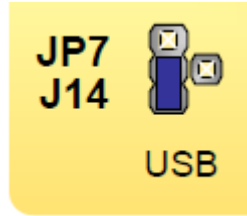
ZyBo Development Board

- The ZyBo provides a **125 MHz clock** to the PL (Programmable Logic) on L16 pin.

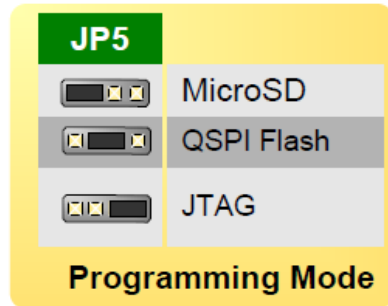


ZyBo Development Board

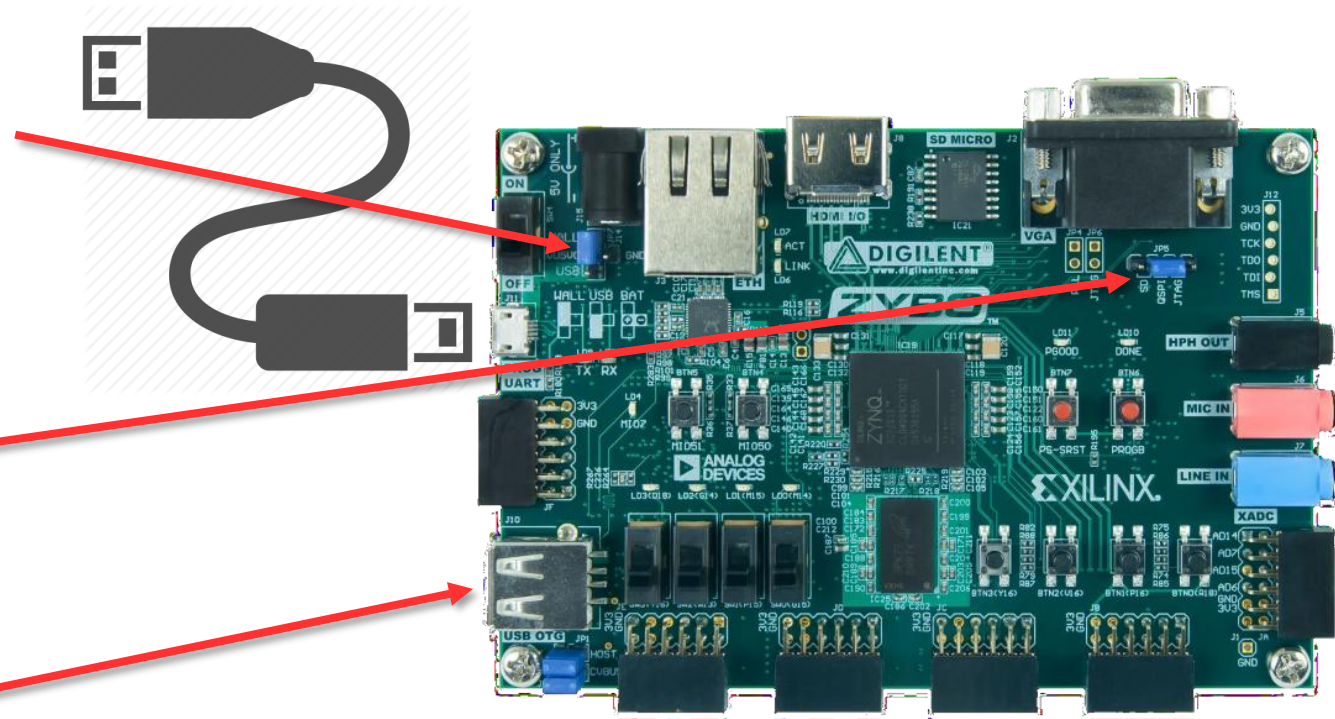
Supply by USB



Programming Mode QSPI



USB: host



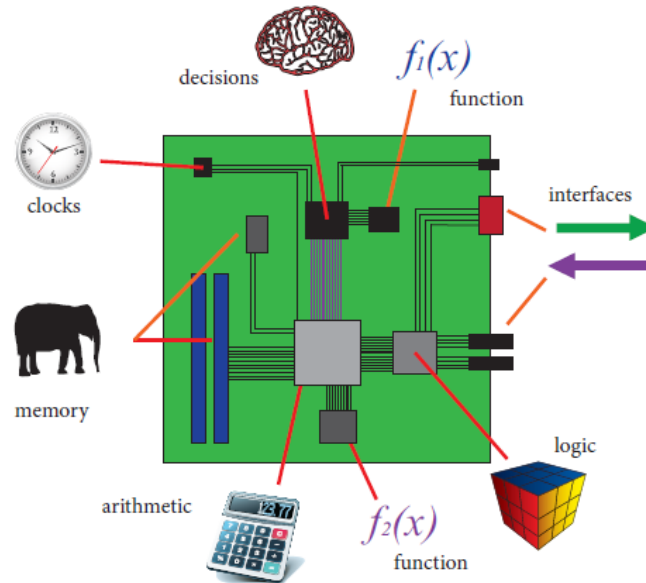
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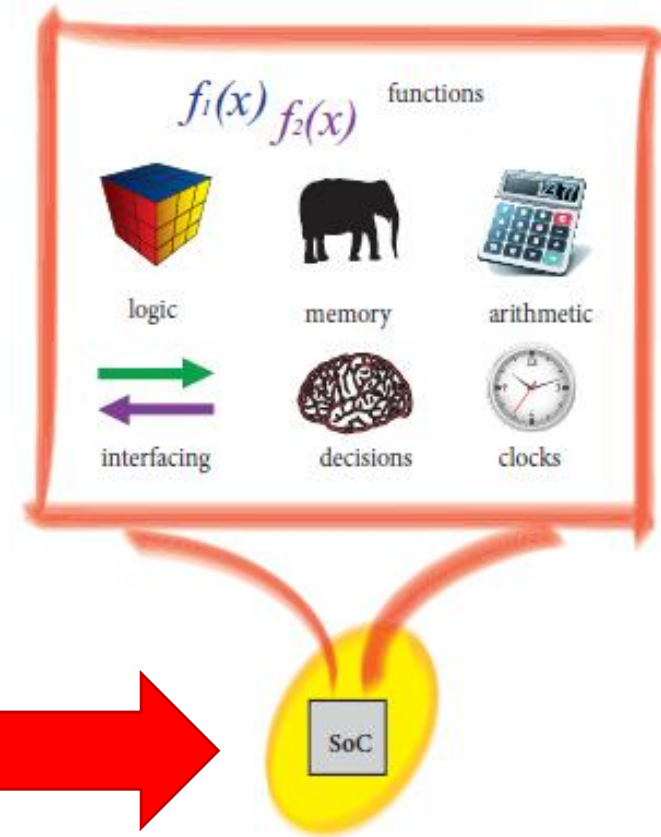
Integration Evolution: SoB $\rightarrow$ SoC

SoC integrates in a single chip:

- Processor/s
- DSPs
- PLDs
- FPGAs
- Interfaces
- Memories
- IPs



System on Board

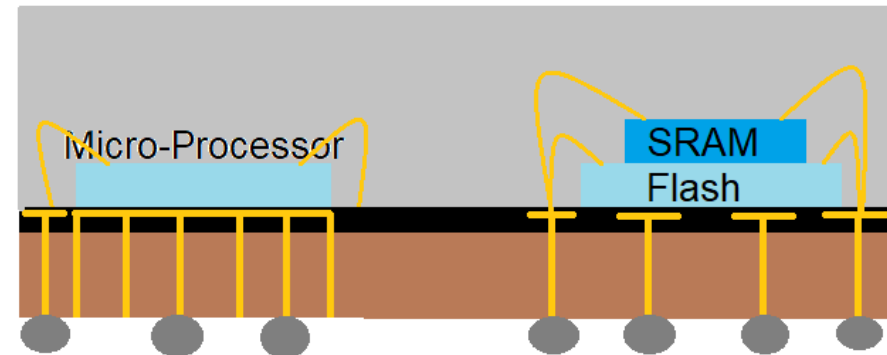


System on Chip (SoC)

Integration Evolution: SoC $\rightarrow$ SiP

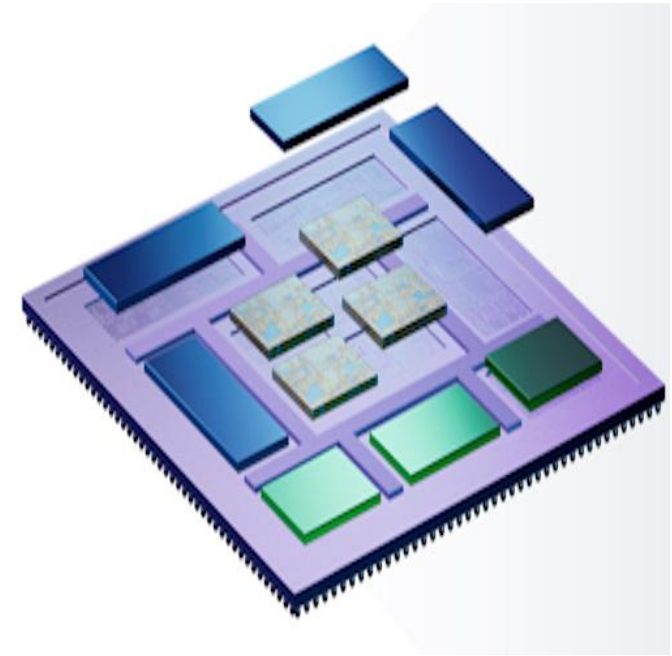
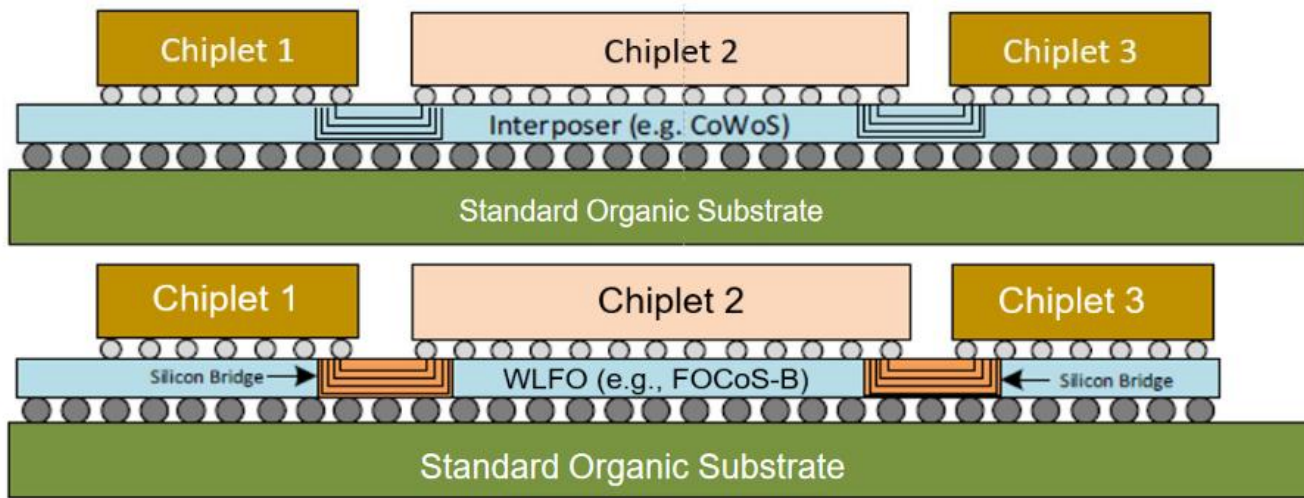


System on Chip (SoC)



System in Package (SiP)

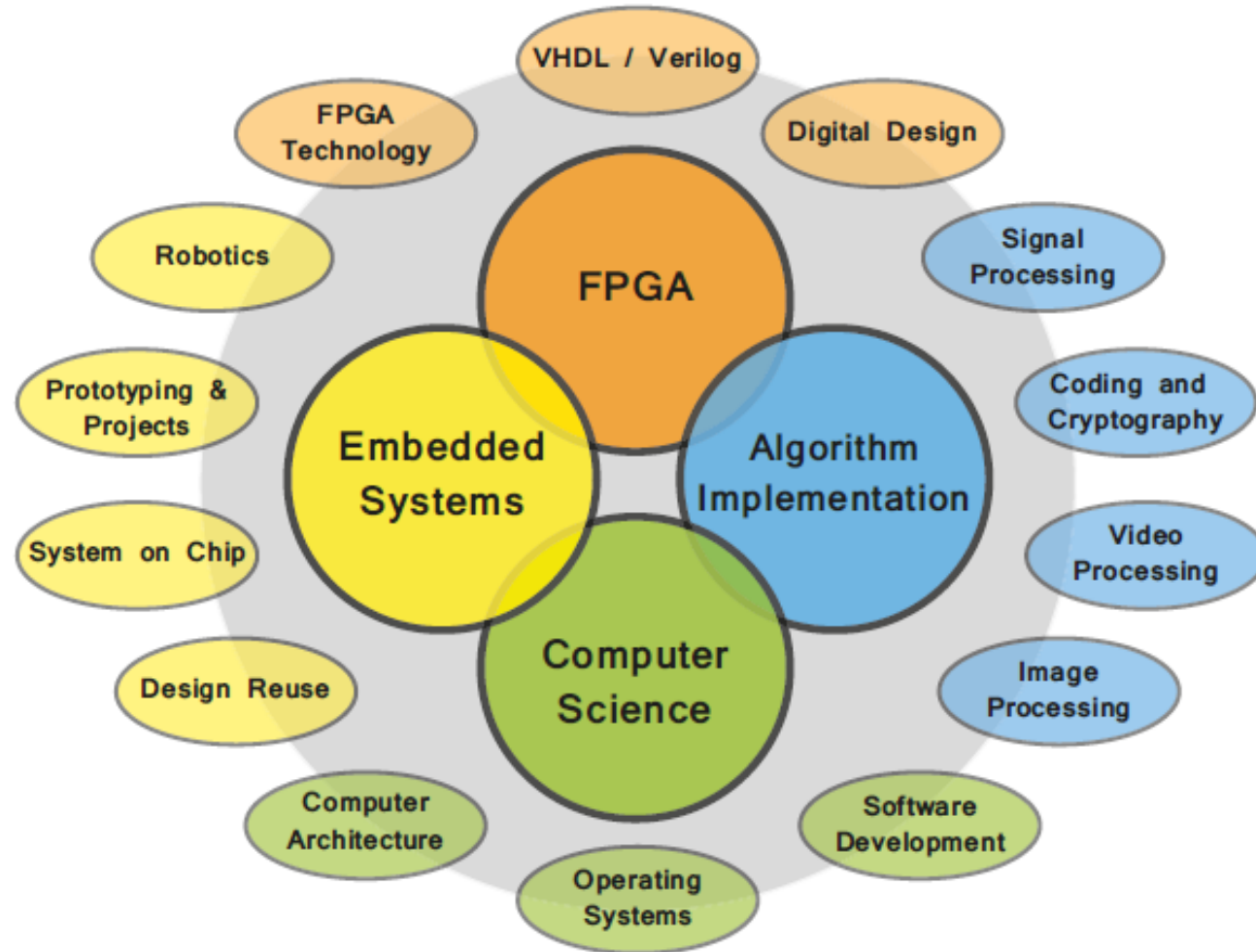
Integration Evolution: Chiplet



Zynq 7000 APSoC

Microelectronic Scopes:

- Medical
- Military and Avionics
- Industrial
- Consumer
- Transport and Mobility
- Telecommunication
- Sensors Interface and IoT

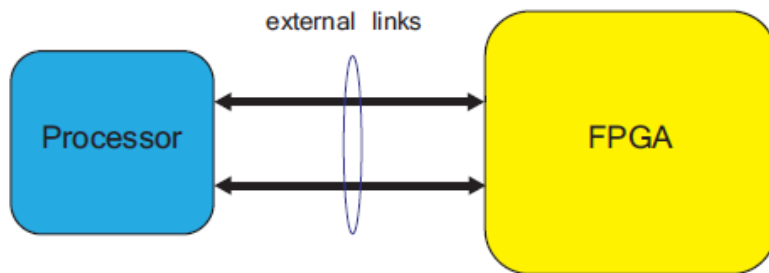


Zynq 7000 APSoC

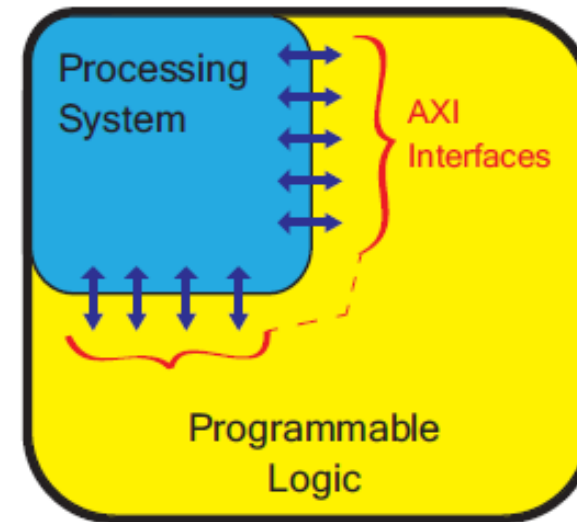
From discrete component to SoC

Xilinx APSoC (All Programmable SoC)

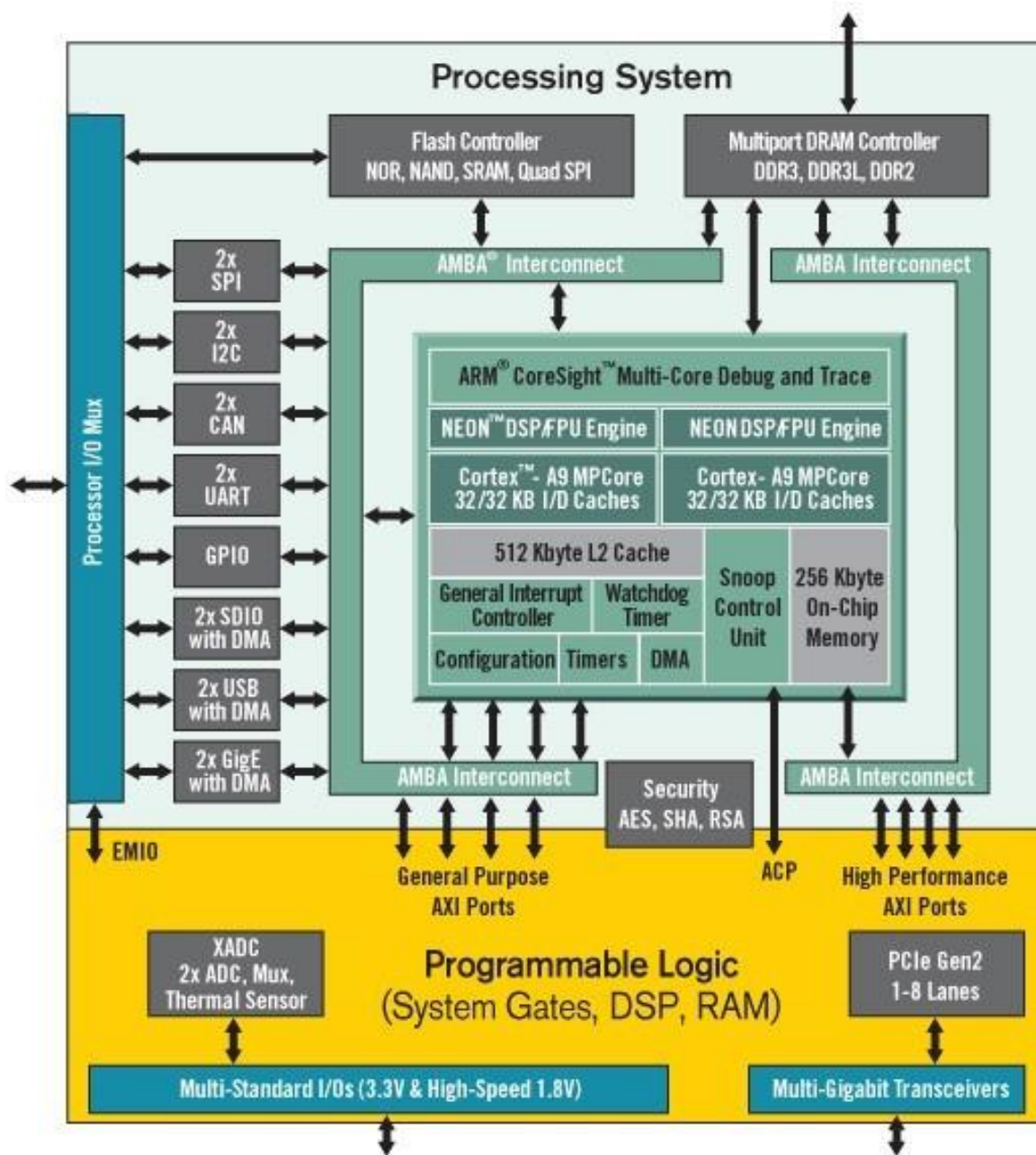
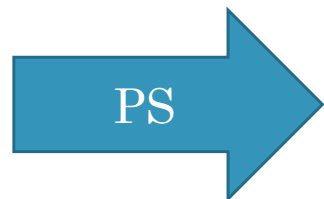
- Single chip as HW/SW design platform
 - PS: Processing System (2x ARM Cortex A-9)
 - PL: Programmable Logic (FPGA 7-series 28nm)



**Discrete component
Architecture**



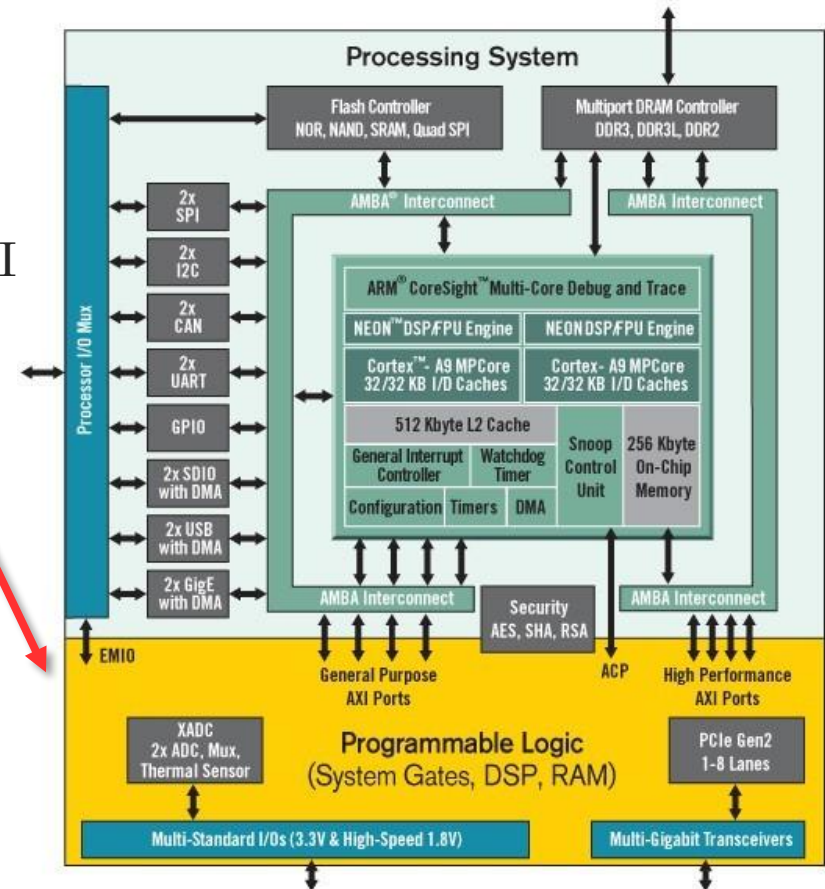
ZYNQ APSoC Architecture



Zynq 7000 APSoC

ZYNQ XC7Z010-1CLG400C

- 650Mhz dual-core Cortex-A9 processor
- DDR3 memory controller with 8 DMA channels
- High-bandwidth peripheral controllers: 1G Eth, USB 2.0, SDI
- Low-bandwidth peripheral controller: SPI, UART, CAN, I2C
- **Reprogrammable logic equivalent to Artix-7 FPGA**
 - 4,400 logic slices, each with four 6-input LUTs and 8 flip-flops
 - 240 KB of fast block RAM
 - Two clock management tiles,
 - each with a phase-locked loop (PLL)
 - and mixed-mode clock manager (MMCM)
 - 80 DSP slices
 - Internal clock speeds exceeding 450MHz
 - On-chip analog-to-digital converter (XADC)

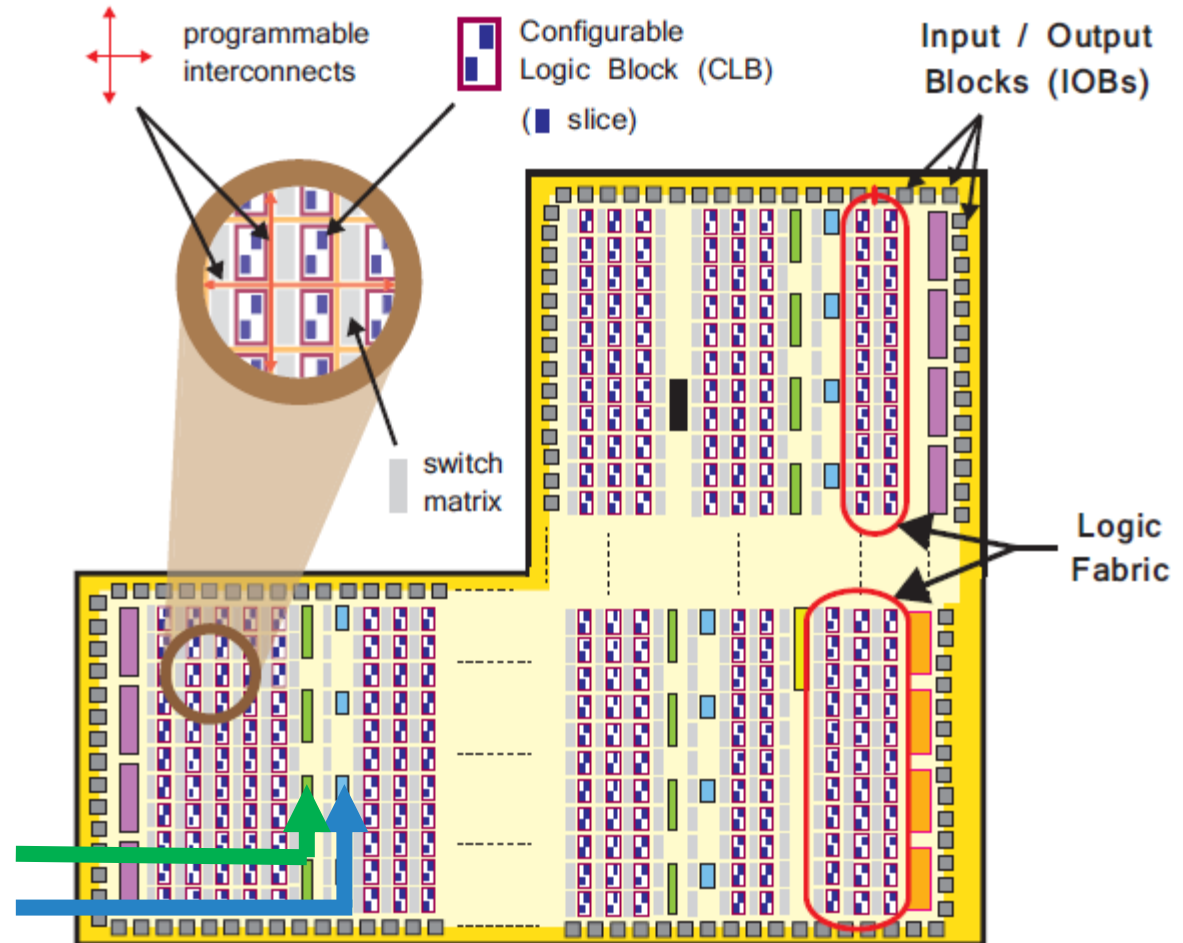


Zynq 7000 APSoC

PL resources:

- Configurable Logic Block (CLB)
 - Slice (x2)
 - LUT (x4)
 - FF (x8)
- Switch Matrix
- Carry Logic
- Input/Output Blocks (IOBs)
- Block RAM
- DSP48E1

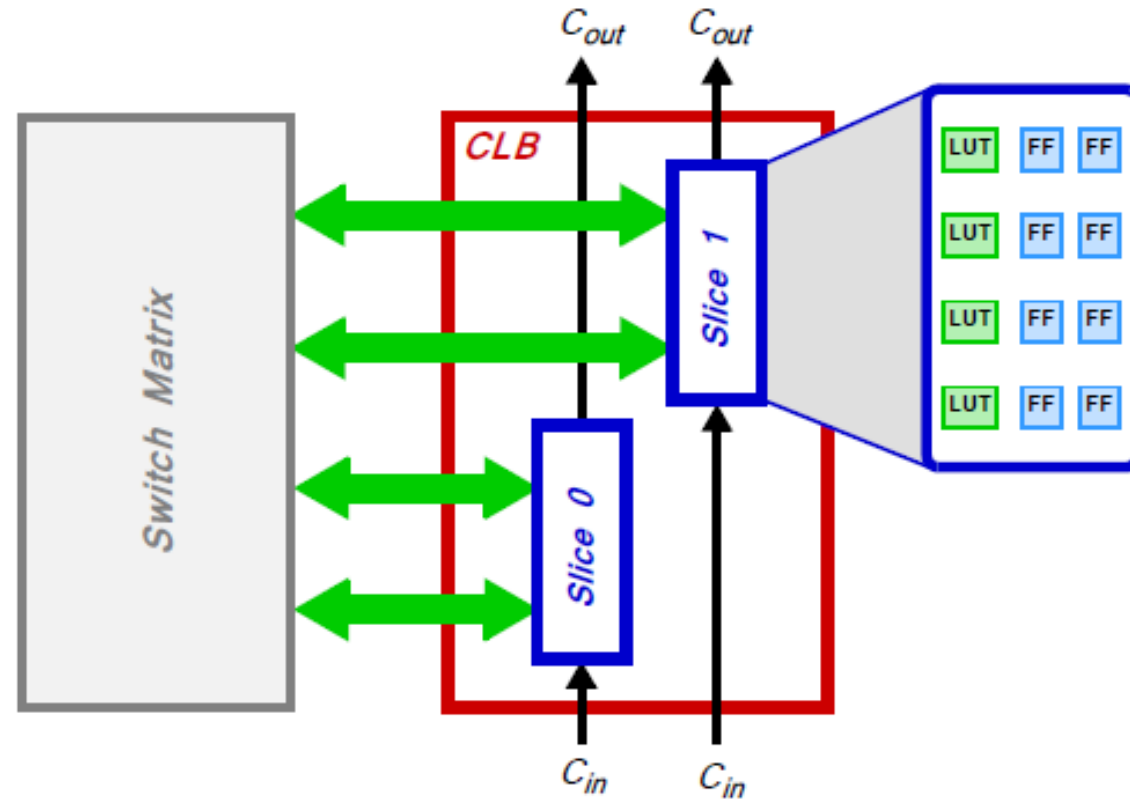
Column of Block RAMs
Column of DSP48E1s



Zynq 7000 APSoC

PL resources:

- **Configurable Logic Block (CLB)**
 - Slice (x2)
 - LUT (x4)
 - FF (x8)
- **Switch Matrix**
- **Carry Logic**
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Zynq 7000 APSoC

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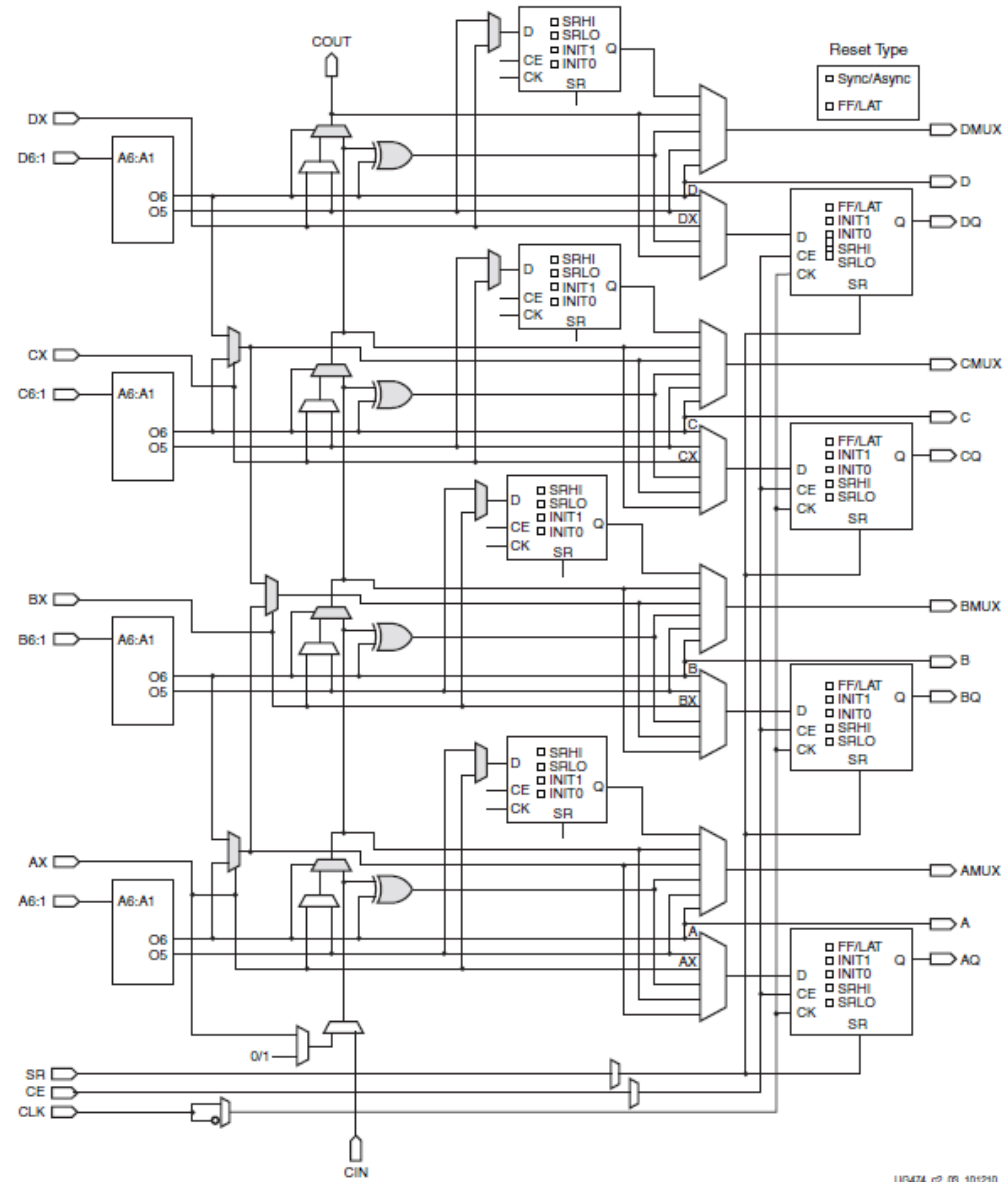


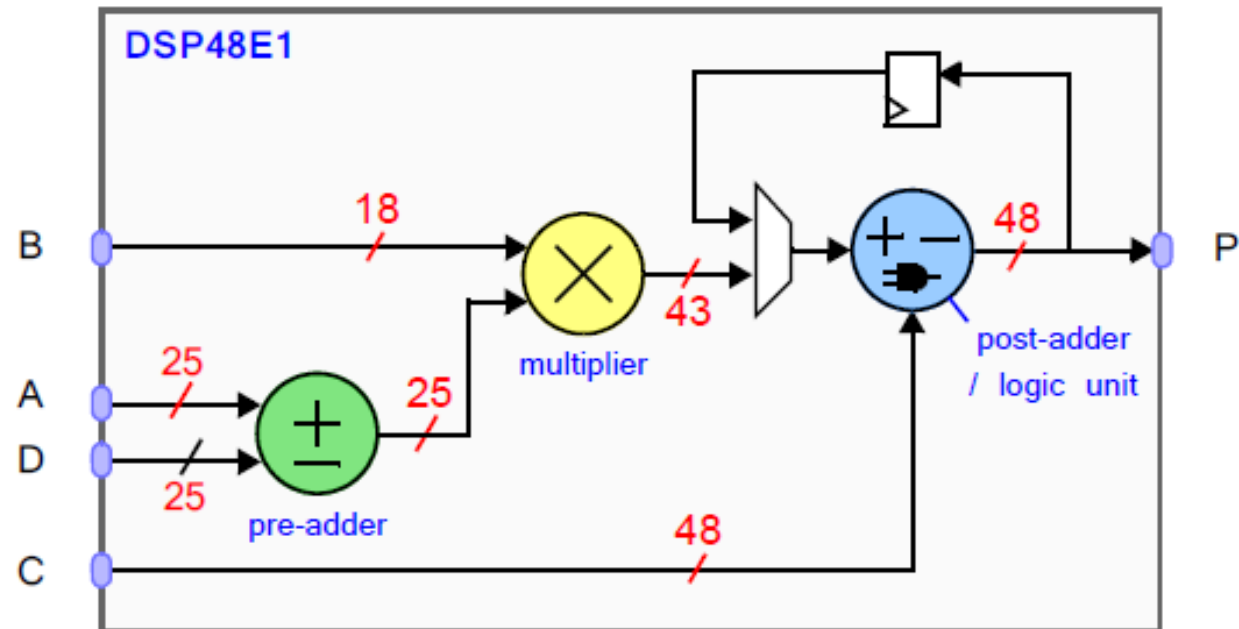
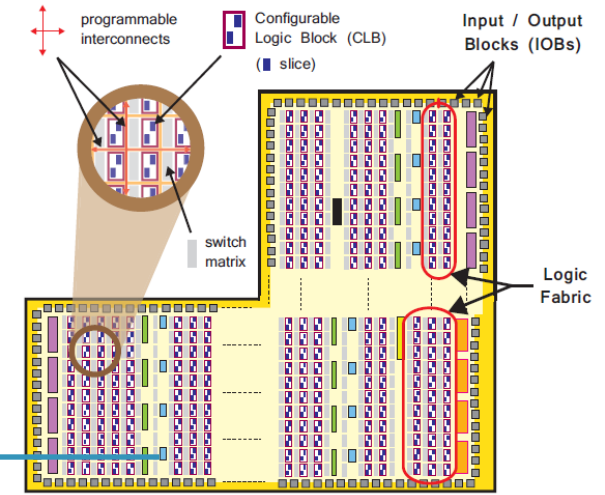
Figure 2-4: Diagram of SLICEL

UG474_v2_03_101210

Zynq 7000 APSoC

PL resources:

- Configurable Logic Block (CLB)
 - Slice (x2)
 - LUT (x4)
 - FF (x8)
- Switch Matrix
- Carry Logic
- Input/Output Blocks (IOBs)
- Block RAM
- **DSP48E1**

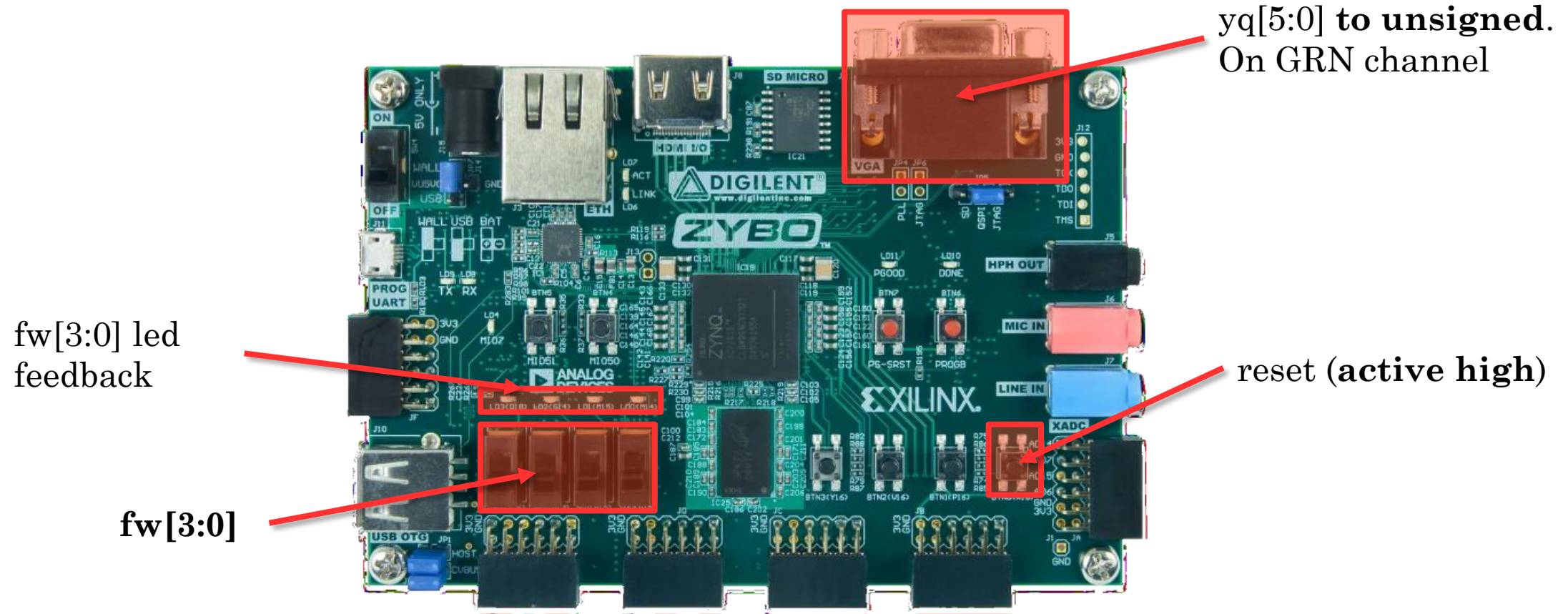


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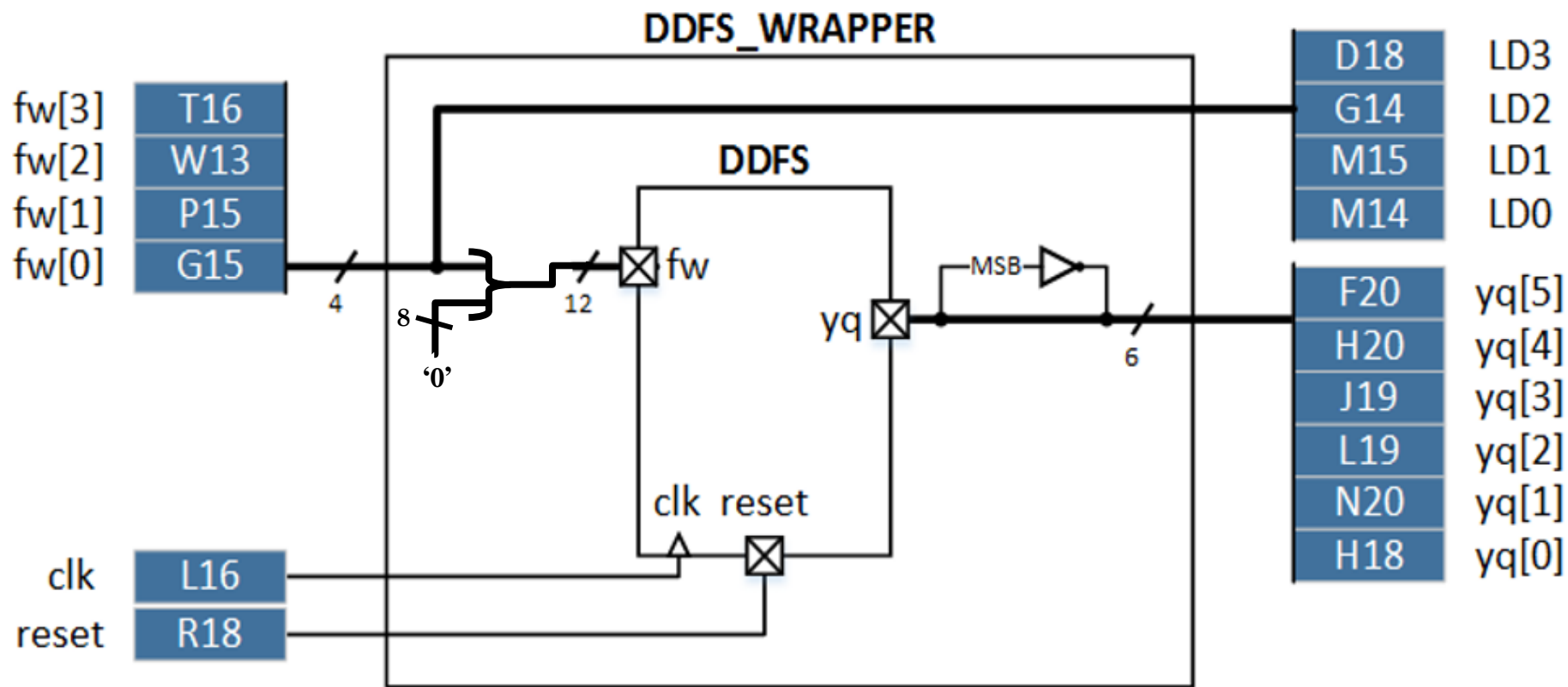
DDFS Wrapper for ZyBo

We need to remap our system into the I/O resources of ZyBo



DDFS Wrapper for ZyBo

Write the wrapper for implementation

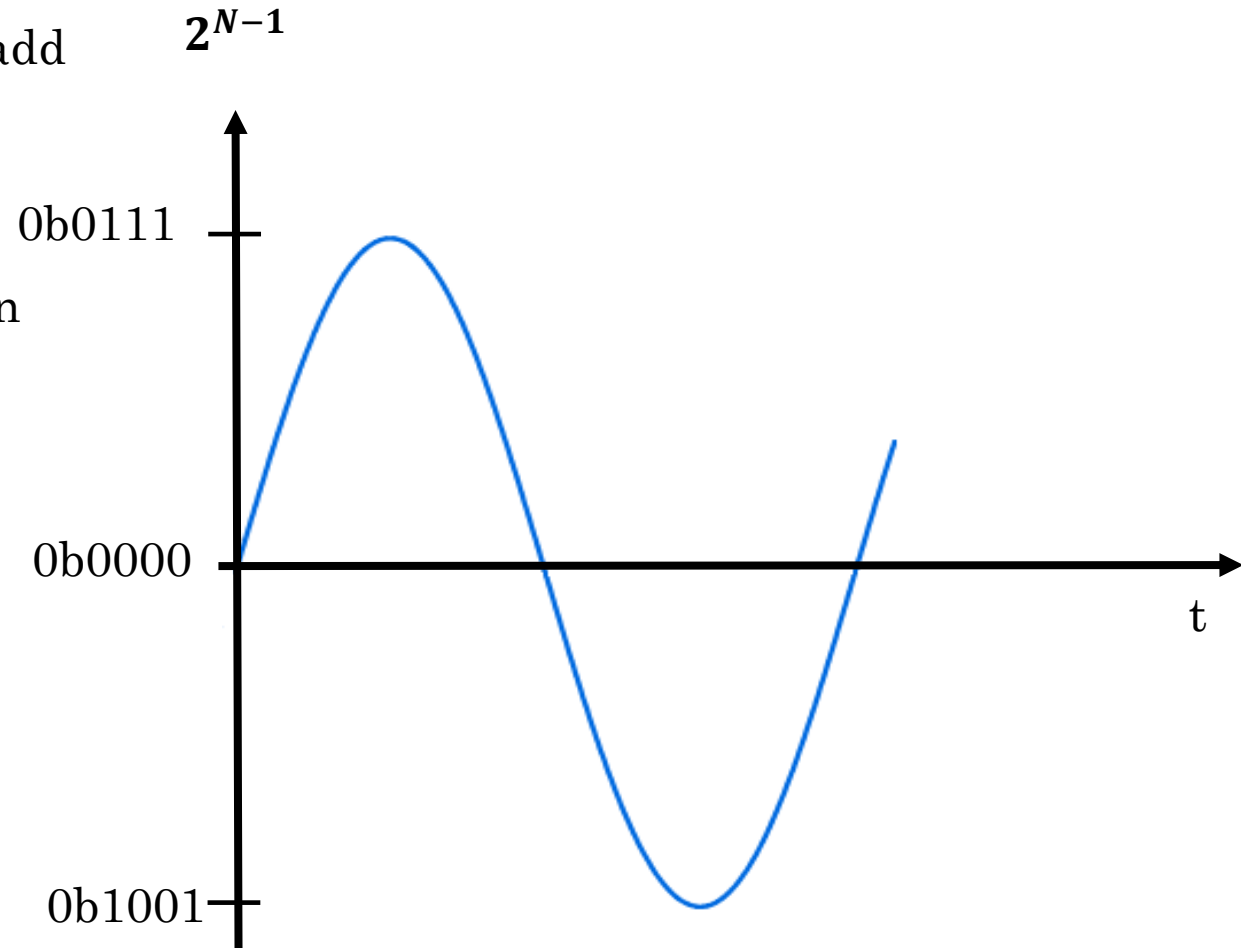


Negation of the MSB: motivation

The DAC converts only to positive values!

The negation of the MSB is equivalent to add 2^{N-1}

Let's suppose to represent *sin* amplitude in C2 using 4 bits



Negation of the MSB: motivation

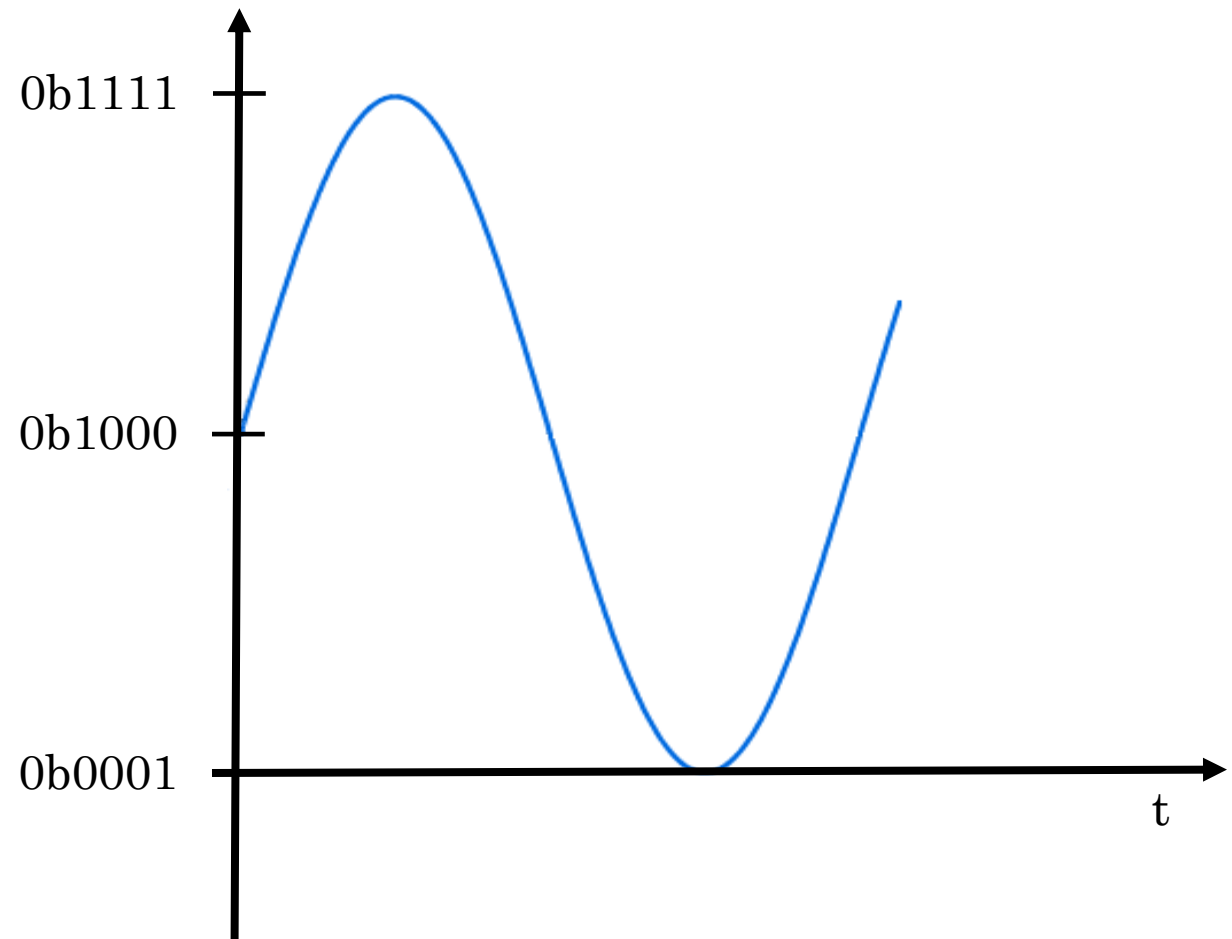
If $N = 4$, 2^{N-1} is equivalent to **0b1000** (unsigned binary representation) and we have that:

- **0b0000 + 0b1000 = 0b1000**
- **0b0111 + 0b1000 = 0b1111**

Same consideration for negative values:

- **0b1001 + 0b1000 = 0b0001**

TRANSLATION!



Negation of the MSB: motivation

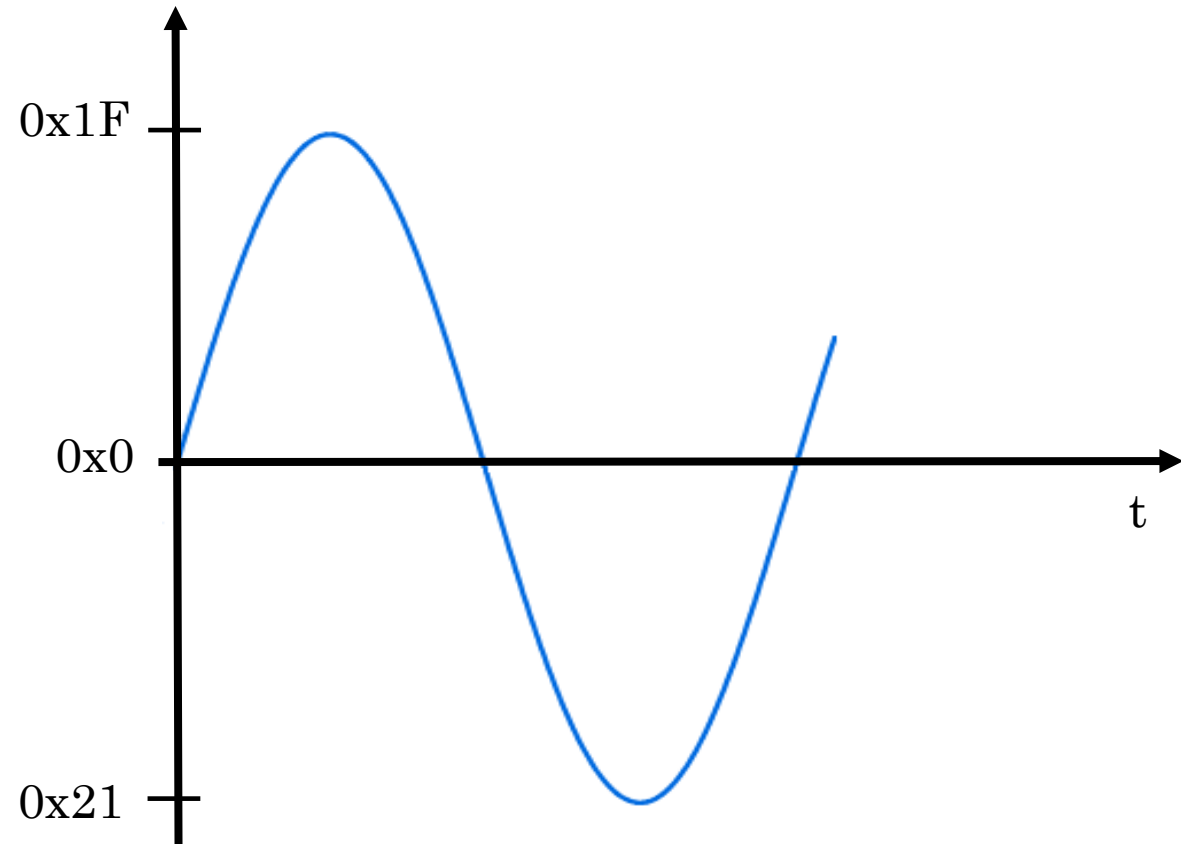
For the DDFS (6 bits):

- Max sample: 31 $\Rightarrow$ 0x1F
- Min sample: -31 $\Rightarrow$ 0x21

Remind: -31 in C2:

- Starting from 31: 0b011111
- Invert all bits: 0b100000
- Add 1: 0b100001

↓
0x21



Negation of the MSB: motivation

- For the DDFS: ($N=6$), 2^{N-1} is equivalent to 0b100000 and we have:

Max sample:

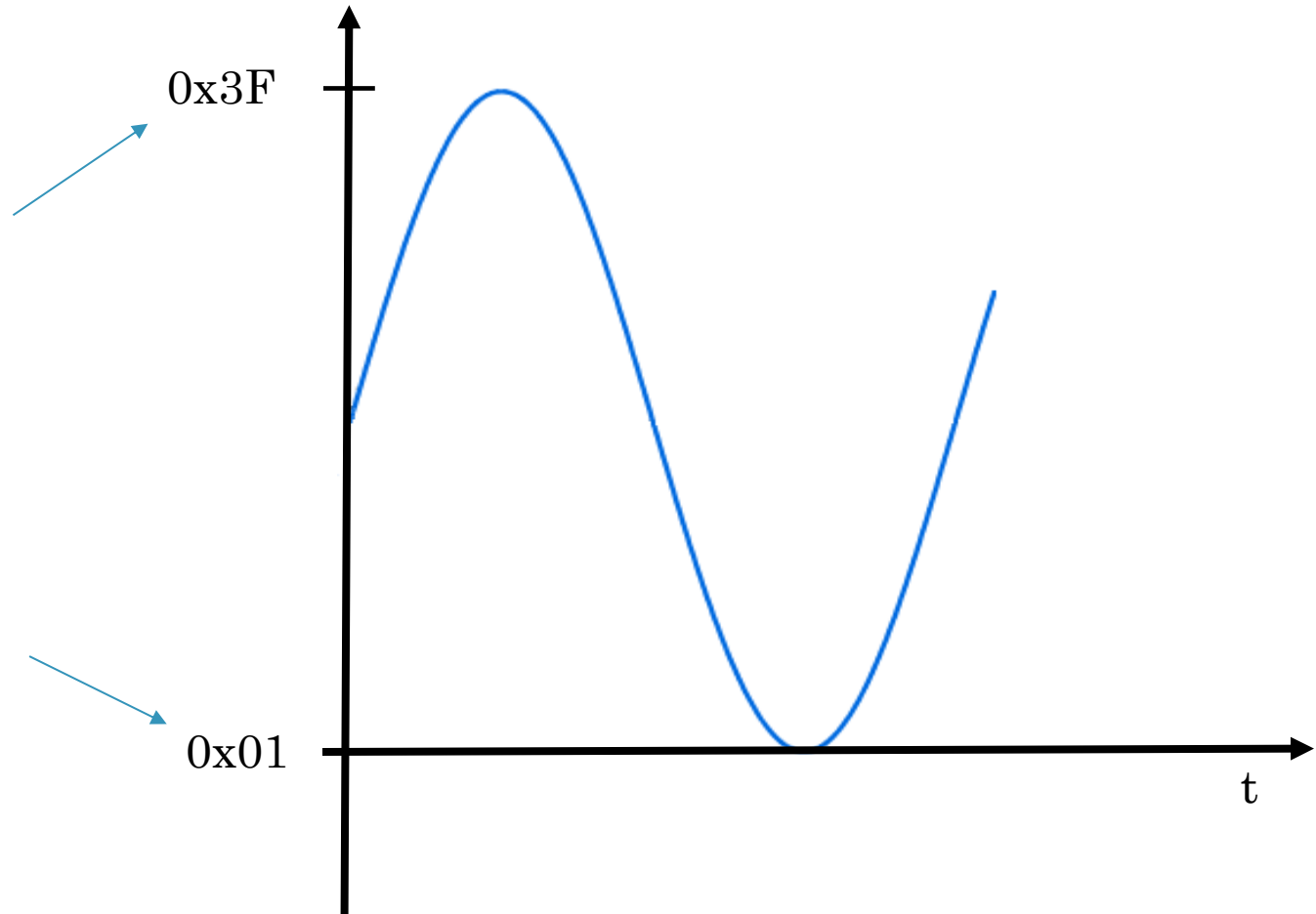
- 0x1F $\Rightarrow$ 0b011111
- 0b011111 + 0b100000 = 0b111111**

0x3F

Min sample:

- 0x21 $\Rightarrow$ 0b100001
- 0b100001 + 0b100000 = 0b000001**

0x01

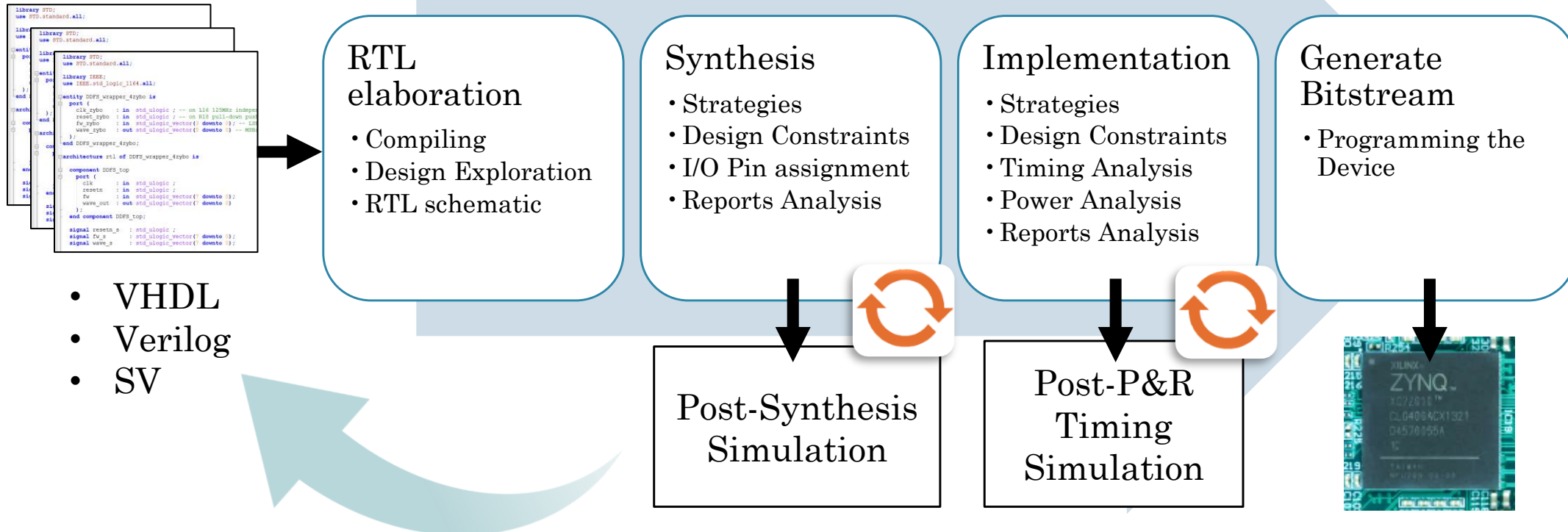


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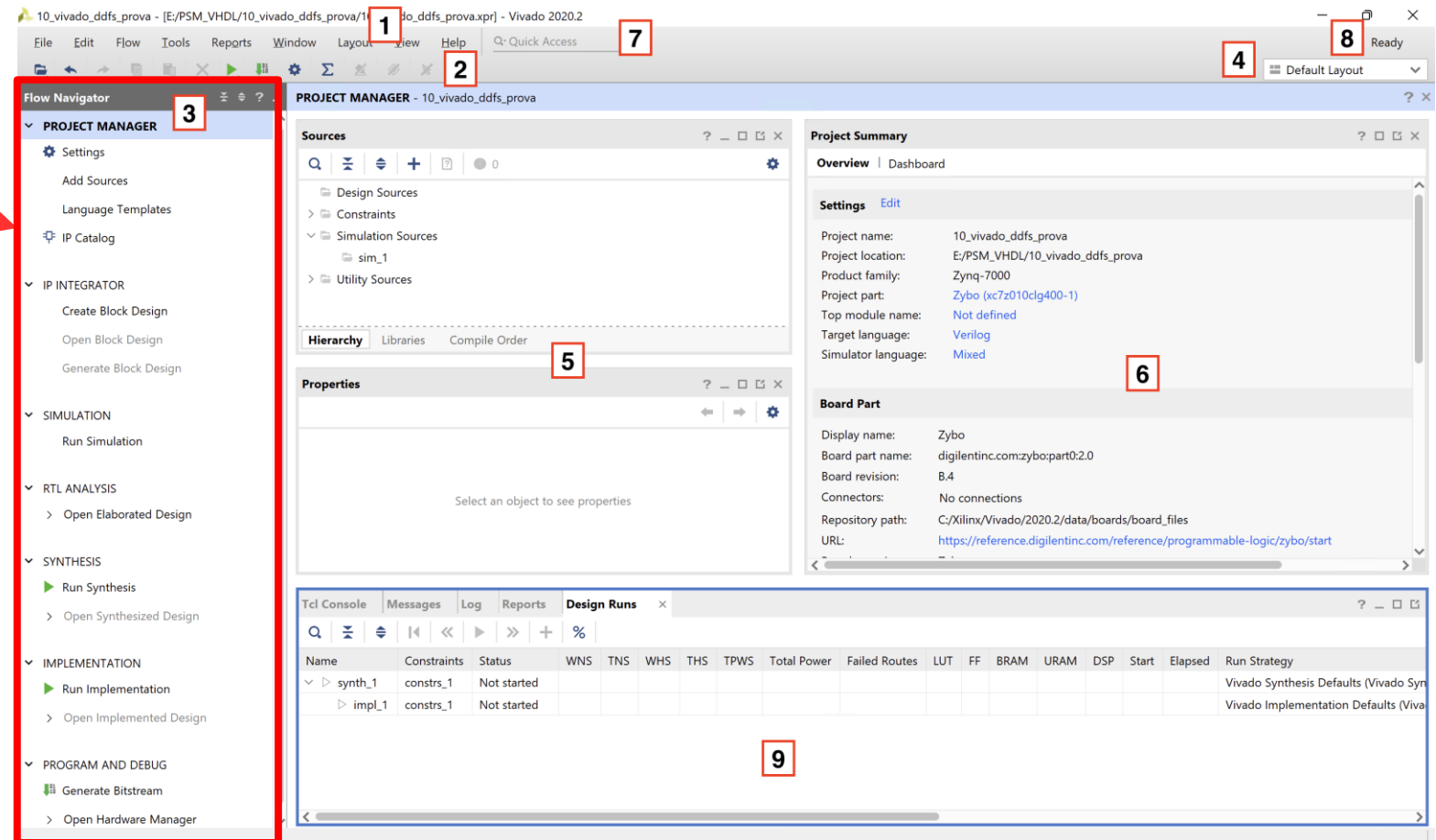
Xilinx VIVADO Design Suite

FPGA Design Flow



Xilinx VIVADO Design Suite

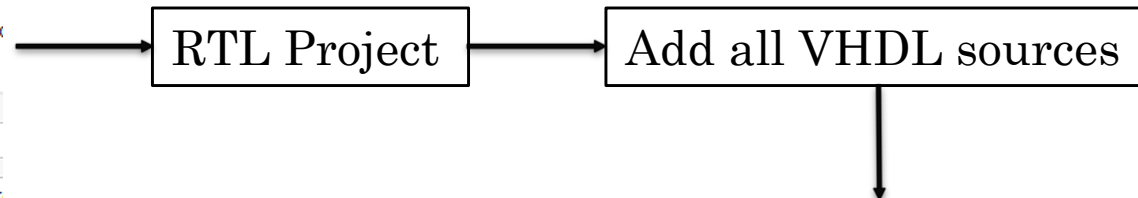
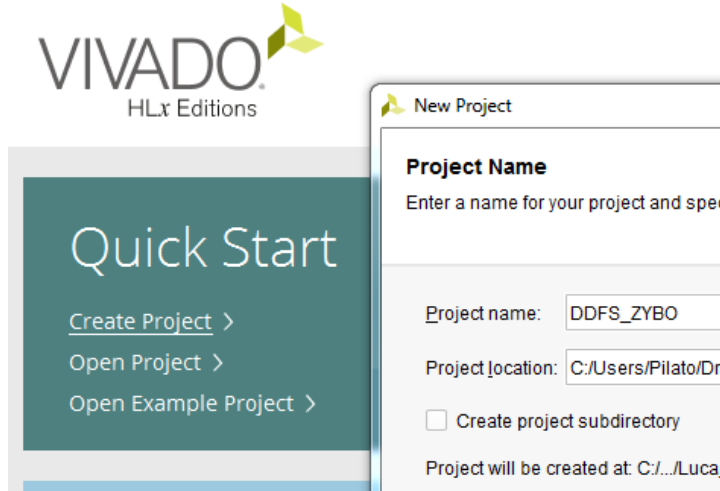
1. Menu Bar
2. Main Toolbar
3. Flow Navigator
4. Layout Selector
5. Data Windows Area
6. Workspace
7. Menu Command Search
8. Project Status Bar
9. Status Bar
10. Result Windows Area



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VIVADO Flow



Parts | Boards

[Reset All Filters](#)

Category: General Purpose Package: clg400 Temperature: All Remaining

Family: Zynq-7000 Speed: -1

Search: Q-

Part	I/O Pin Count	Available IOBs	LUT Elements	FlipFlops	Block RAMs	Ultra RAMs	DSPs
xc7z007sclg400-1	400	100	14400	28800	50	0	66
xc7z010clg400-1	400	100	17600	35200	60	0	80
xc7z014sclg400-1	400	125	40600	81200	107	0	170
xc7z020clg400-1	400	125	53200	106400	140	0	220

Device:
xc7z010clg400-1

VIVADO Flow



New Project

Add Sources

Specify HDL, netlist, Block Design, and IP files, or directories containing those files, to add to your project. Create a new source file on disk and add it to your project. You can also add and create sources later.

	Index	Name	Library	HDL Source For	Location
	1	Counter.vhd	work	Synthesis & Simulation	C:/Users/emilio/UNI/PHD/Supporto alla did
	2	DFF_N.vhd	work	Synthesis & Simulation	C:/Users/emilio/UNI/PHD/Supporto alla did
	3	ddfs.vhd	work	Synthesis & Simulation	C:/Users/emilio/UNI/PHD/Supporto alla did
	4	ddfs_lut_4096.vhd	work	Synthesis & Simulation	C:/Users/emilio/UNI/PHD/Supporto alla did
	5	ddfs_wrapper.vhd	work	Synthesis & Simulation	C:/Users/emilio/UNI/PHD/Supporto alla did
	6	full_adder.vhd	work	Synthesis & Simulation	C:/Users/emilio/UNI/PHD/Supporto alla did
	7	ripple_carry_adder.vhd	work	Synthesis & Simulation	C:/Users/emilio/UNI/PHD/Supporto alla did

Add Files

Add Directories

Create File

☐ Scan and add RTL include files into project

☐ Copy sources into project

☒ Add sources from subdirectories

Target language: VHDL

Simulator language: Mixed



< Back

Next >

Finish

Cancel

When loading src files, select the «work» library if you have packages to include

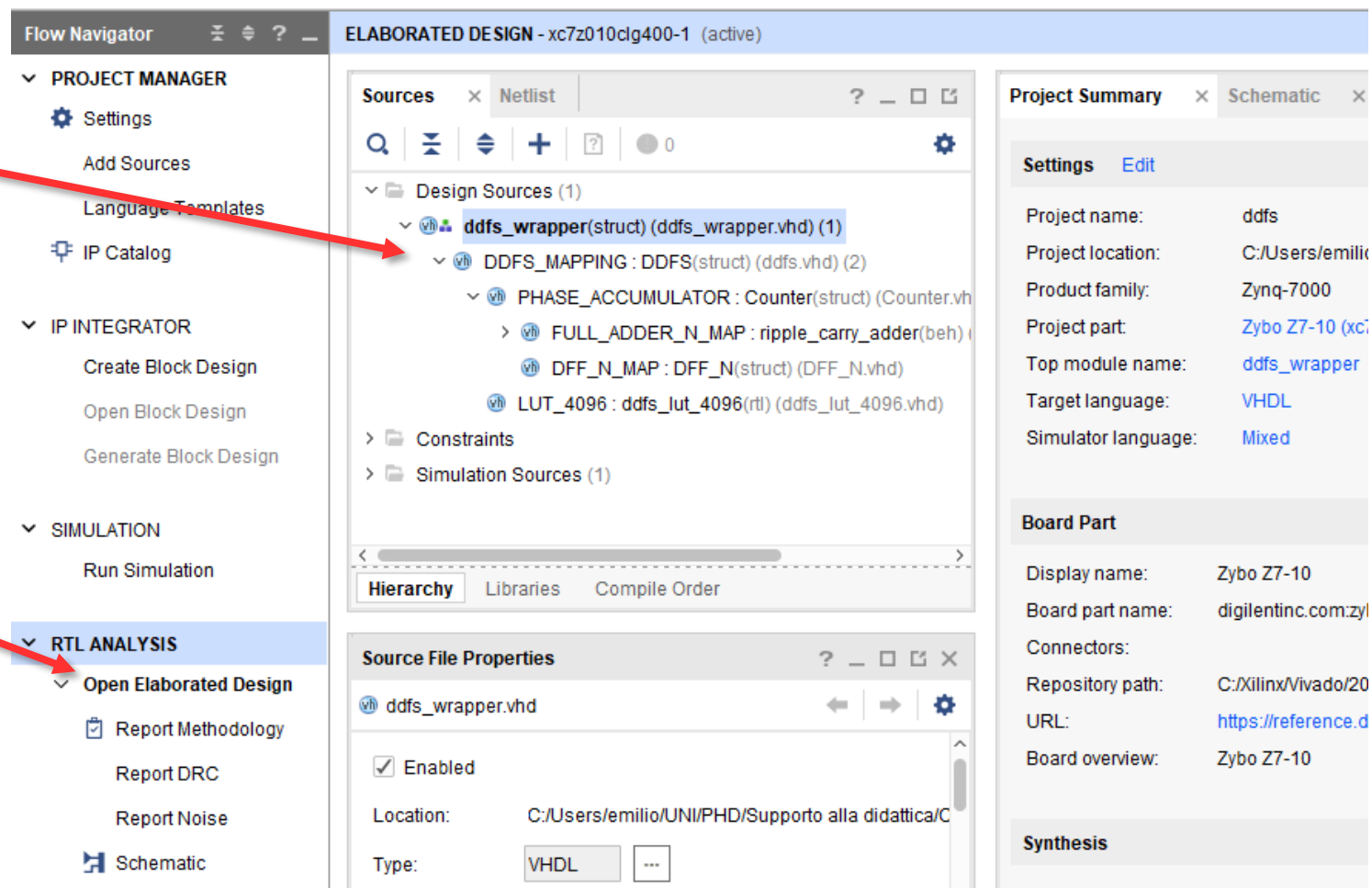
VIVADO Flow



Hierarchy is automatically resolved

Open Elaborated Design:

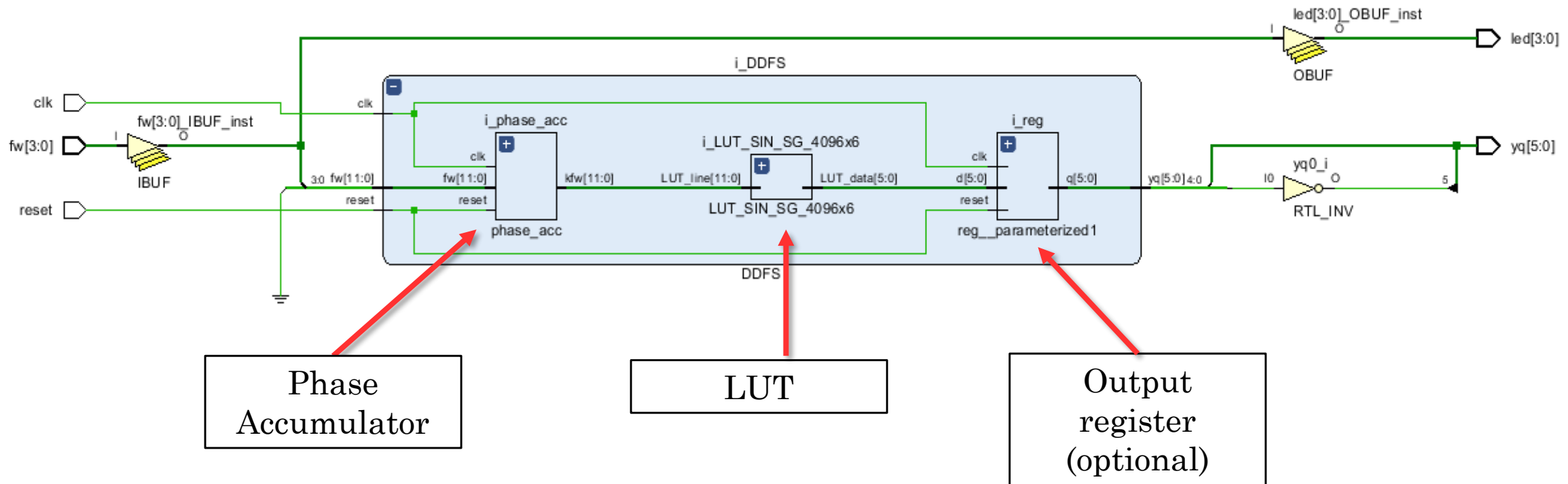
- Checks VHDL consistency
- Allows RTL analysis



VIVADO Flow



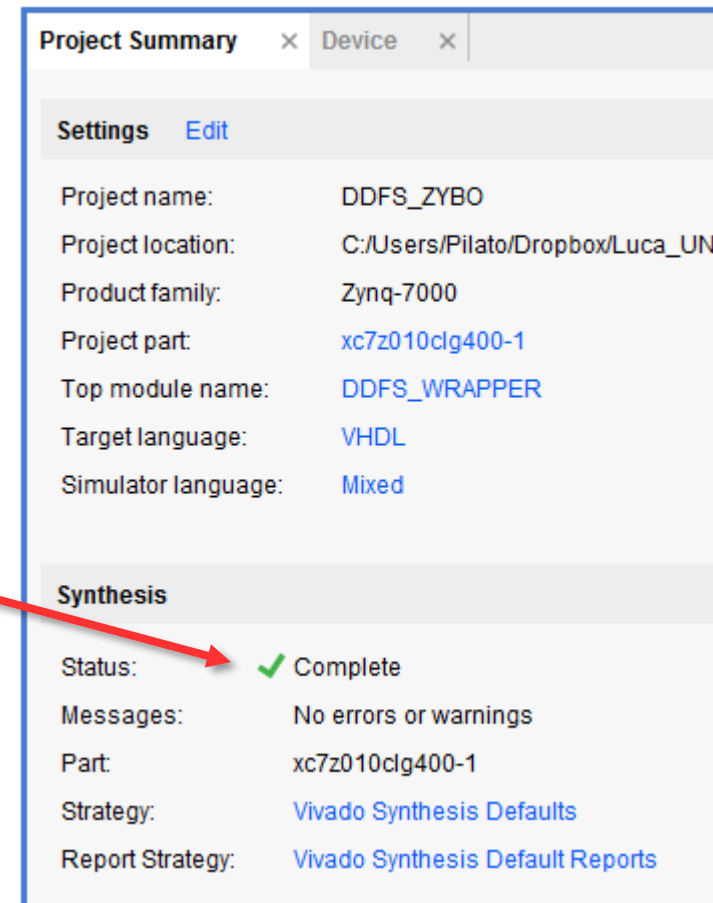
RTL ANALYSIS → Open Elaborated Design → Schematic



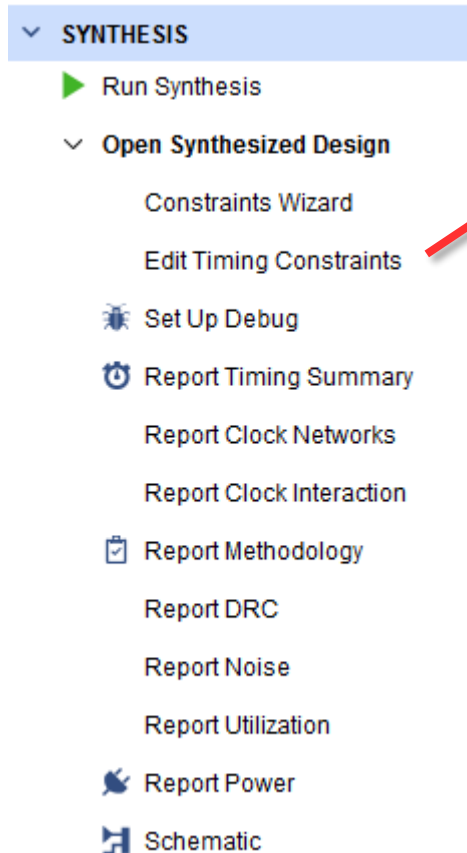
VIVADO Flow



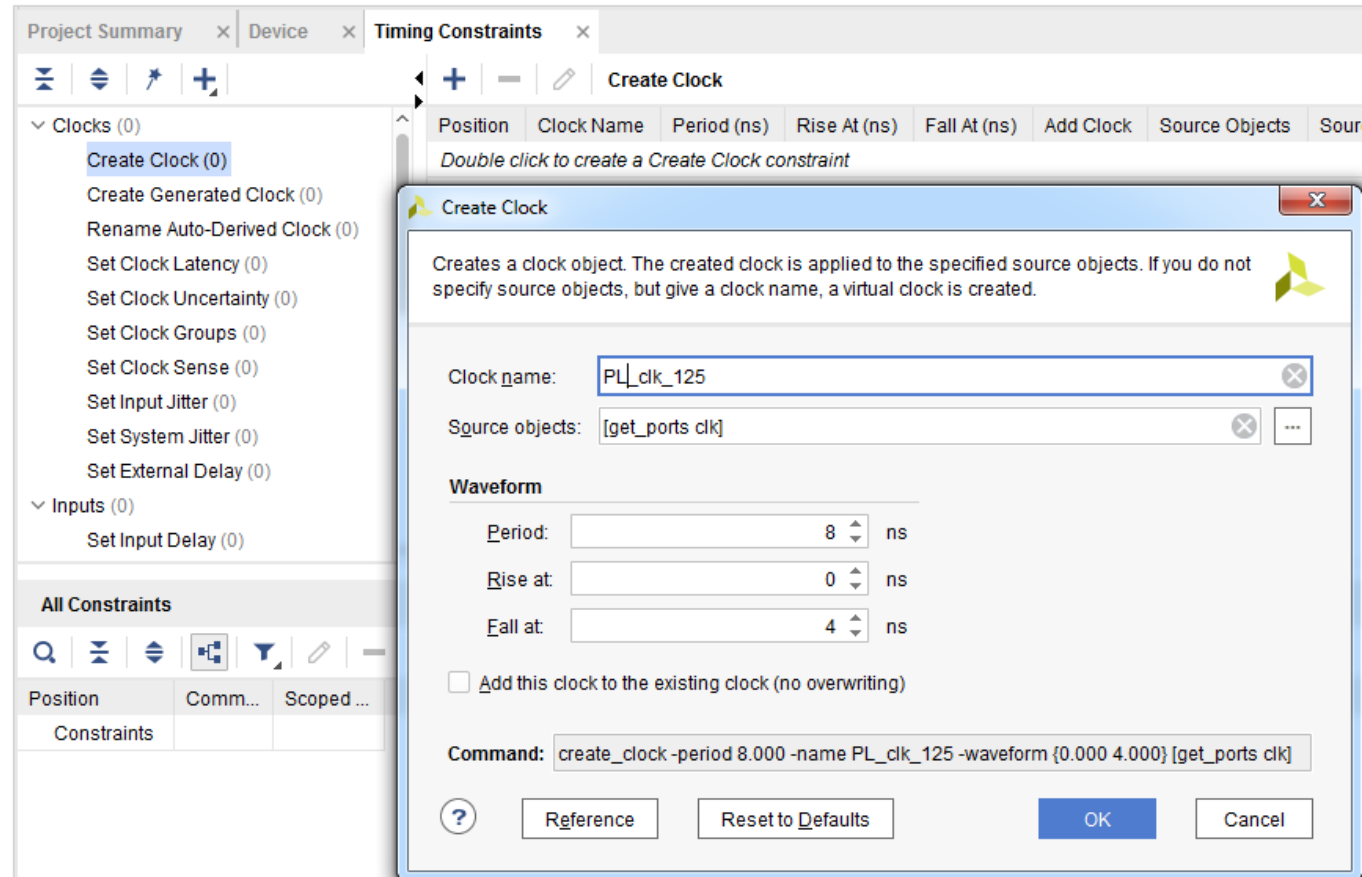
- Close the Elaborated Design
- Run a first Synthesis (without constraints)
- ...
- ...
- At the end Open Synthesized Design
- Open Project Summary
- Check any messages



VIVADO Flow



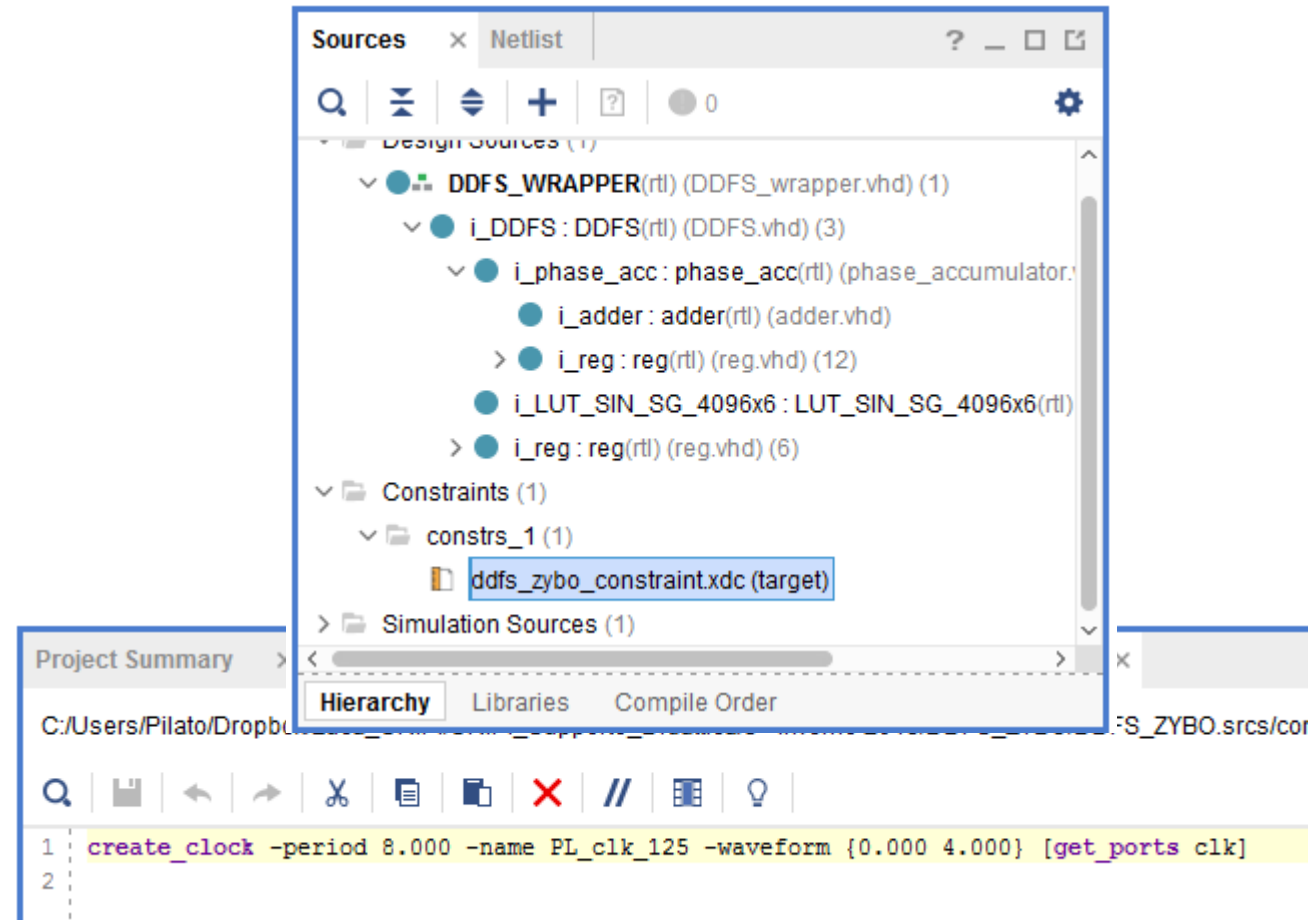
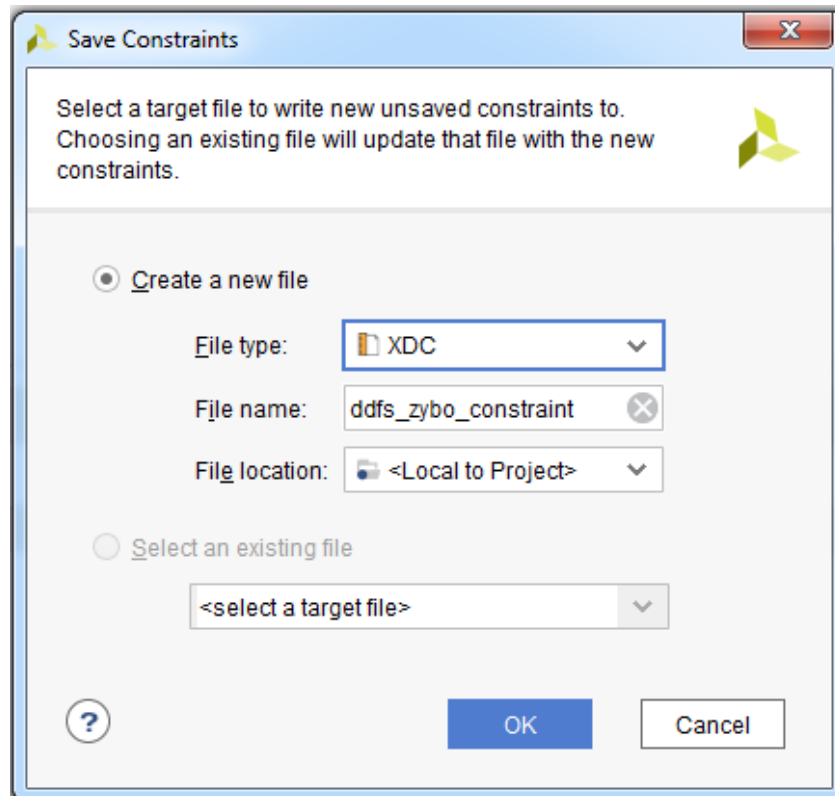
We need to specify clock constraints



VIVADO Flow



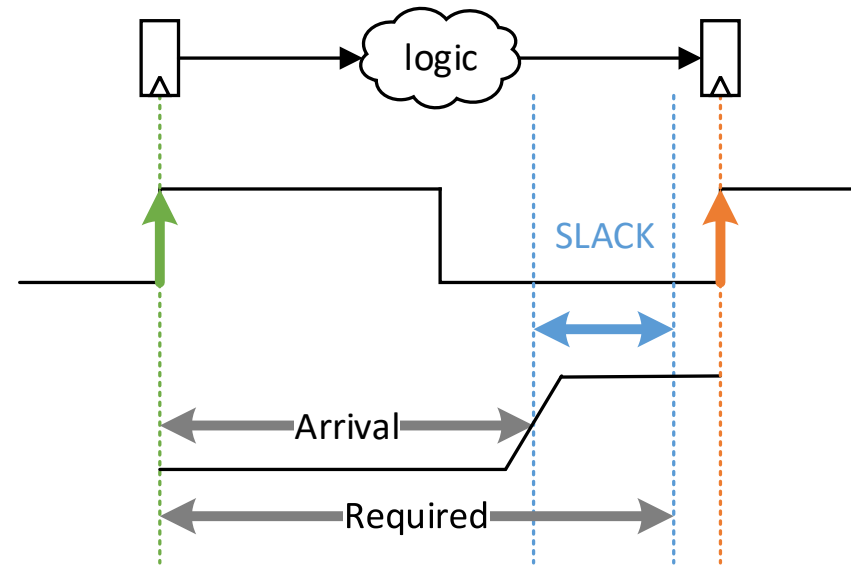
Save and look the xdc file



VIVADO Flow



- Re-Run the synthesis
- Check the Slack (Report Timing Summary)
 - Slack: “Arrival – Required” Time
 - Positive OK
 - Negative BAD



WNS: Worst Negative Slack (CRITICAL PATH)
TNS: Total Negative Slack (sum of negative slacks)

Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 3,196 ns	Worst Hold Slack (WHS): 0,170 ns	Worst Pulse Width Slack (WPWS): 3,500 ns
Total Negative Slack (TNS): 0,000 ns	Total Hold Slack (THS): 0,000 ns	Total Pulse Width Negative Slack (TPWS): 0,000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 18	Total Number of Endpoints: 18	Total Number of Endpoints: 19

All user specified timing constraints are met.

VIVADO Flow



- Write netlist to file for:
 - Check primitives
 - Run post-synth simulations (we skip this)

```
Tcl Console x Messages Log Reports Design Runs Package Pins I/O Ports
[Icons]
WARNING: [filemgmt 56-3] Board Part Repository Path: Could not find the directory
WARNING: [filemgmt 56-3] Default IP Output Path : Could not find the directory '
Scanning sources...
Finished scanning sources
INFO: [IP_Flow 19-234] Refreshing IP repositories
INFO: [IP_Flow 19-1704] No user IP repositories specified
INFO: [IP_Flow 19-2313] Loaded Vivado IP repository 'C:/Xilinx/Vivado/2020.2/data
open_project: Time (s): cpu = 00:00:07 ; elapsed = 00:00:08 . Memory (MB): peak :
update_compile_order -fileset sources_1
open_run synth_1 -name synth_1
write_vhdl -mode funcsim ./ddfs_synt.vhd
```

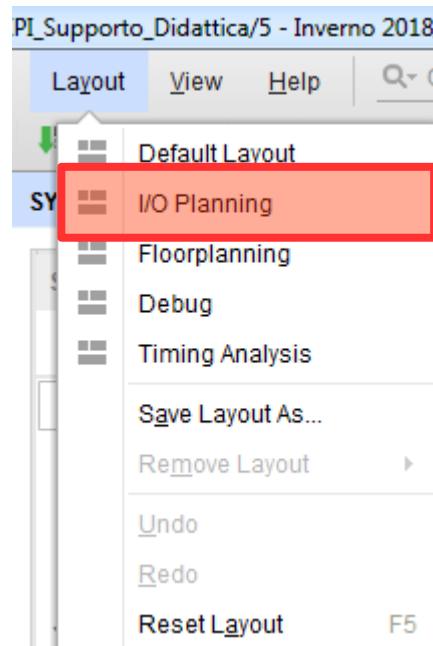
```
use IEEE.STD_LOGIC_1164.ALL;
library UNISIM;
use UNISIM.VCOMPONENTS.ALL;
entity DFF_N is
    port (
        \q_reg_rep[4]_0\ : out STD_LOGIC;
        Q : out STD_LOGIC_VECTOR ( 11 downto 0 );
        CLK : in STD_LOGIC;
        AR : in STD_LOGIC_VECTOR ( 0 to 0 );
        led_out_OBUF : in STD_LOGIC_VECTOR ( 3 downto 0 );
    );
end DFF_N;

architecture STRUCTURE of DFF_N is
    signal \FULL_ADDER_N_i/s13_0\ : STD_LOGIC;
    signal \FULL_ADDER_N_i/s25_4\ : STD_LOGIC;
    signal ^q\ : STD_LOGIC_VECTOR ( 11 downto 0 );
    signal fullAdder_out : STD_LOGIC_VECTOR ( 11 downto 0 );
    signal phase_out : STD_LOGIC_VECTOR ( 11 downto 0 );
    attribute SOFT_HLUTNM : string;
    attribute SOFT_HLUTNM of \q[0]_i_1\ : label is "soft_lutpair18";
    ...
    attribute equivalent_register_removal of \q_reg_rep[9]\ : label is "no";
begin
    Q(11 downto 0) <= ^q(11 downto 0);
    \output_reg[3]_i_10\ : unisim.vcomponents.LUT2
        generic map(
            INIT => X"1"
        )
        port map (
            I0 => ^q(4),
            I1 => ^q(5),
            O => \q_reg_rep[4]_0\
        );
    \q[0]_i_1\ : unisim.vcomponents.LUT2
        generic map(
            INIT => X"6"
        )
        port map (
            I0 => phase_out(0),
            I1 => led_out_OBUF(0),
            O => fullAdder_out(0)
        );
    \q[10]_i_1\ : unisim.vcomponents.LUT5
        generic map(
            INIT => X"6AAAAAAA"
        )
```

VIVADO Flow



It is possible to assign I/O pin before Implementation. (Physical Constraints)



Tcl Console	Messages	Log	Reports	Design Runs	Timing	Package Pins	I/O Ports		
Name	Direction	Neg Diff Pair	Package Pin	Fixed	Bank	I/O Std	Vcco	Vref	
All ports (16)									
fw (4)									
fw[3]	IN		T16	☒	34	LVC MOS33*	3.300		
fw[2]	IN			☐		default (LVC MOS18)	1.800		
fw[1]	IN			☐		default (LVC MOS18)	1.800		
fw[0]	IN			☐		default (LVC MOS18)	1.800		
led (4)									
led	OUT			☐		default (LVC MOS18)	1.800		
yq (6)									
yq	OUT			☐		default (LVC MOS18)	1.800		
Scalar ports (2)									

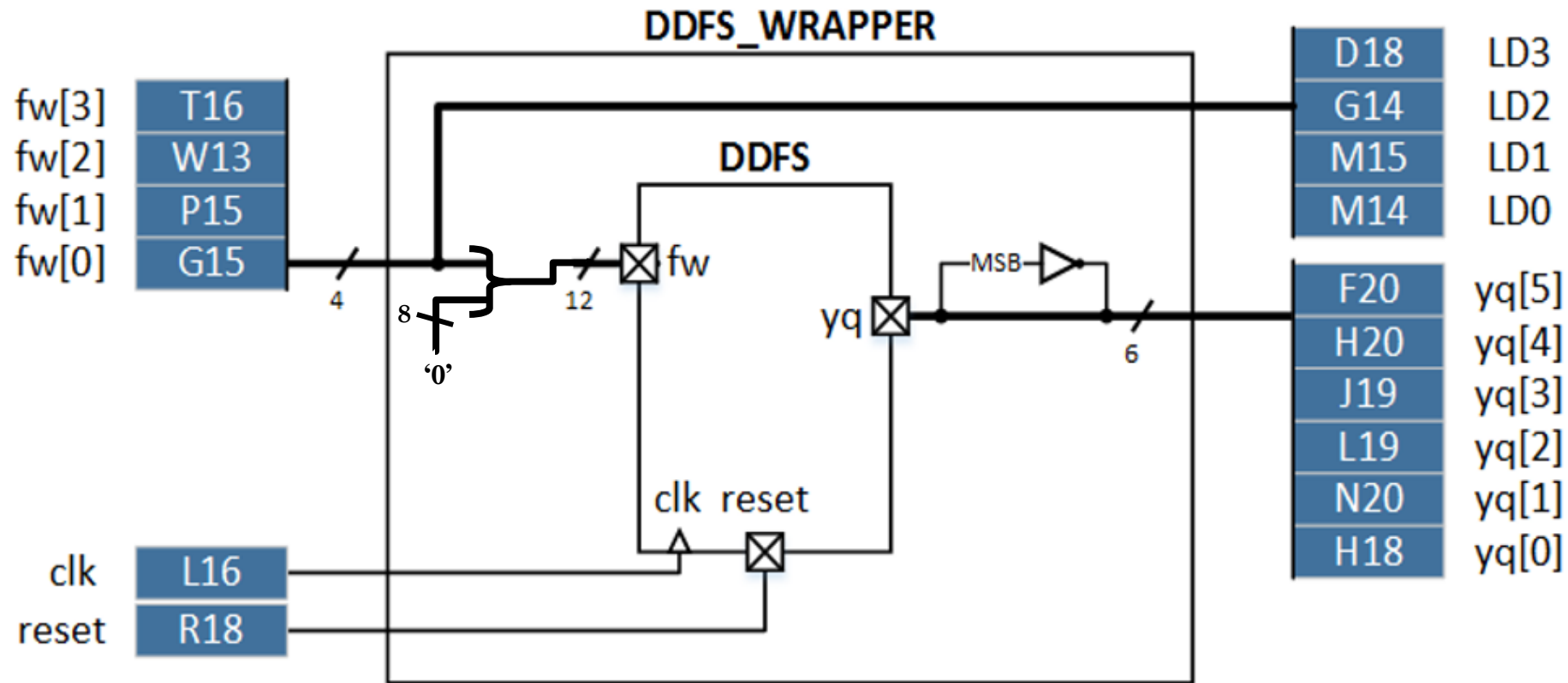
Location
(BOARD)

LVC MOS33
(BOARD I/O supply)

VIVADO Flow



It is possible to assign I/O pin before Implementation. (Physical Constraints)



VIVADO Flow

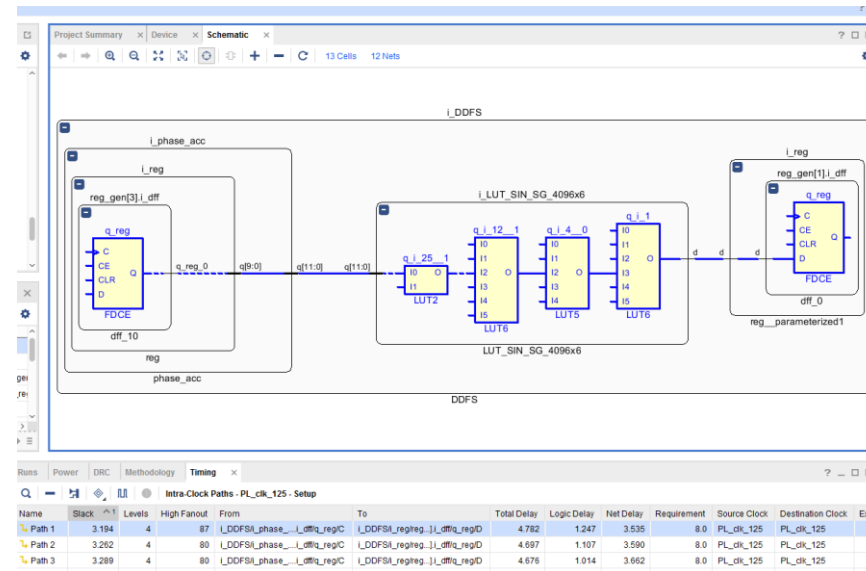


Run an Implementation (with constraints)

...

At the end Open Implemented Design:

- Project Summary
 - Check errors/warnings/messages
- Report timing
 - Check the slack



Design Timing Summary

Setup

Worst Negative Slack (WNS): **3,194 ns**

Total Negative Slack (TNS): 0,000 ns

Number of Failing Endpoints: 0

Total Number of Endpoints: 18

Hold

Worst Hold Slack (WHS): 0,228 ns

Total Hold Slack (THS): 0,000 ns

Number of Failing Endpoints: 0

Total Number of Endpoints: 18

Pulse Width

Worst Pulse Width Slack (WPWS): 3,500 ns

Total Pulse Width Negative Slack (TPWS): 0,000 ns

Number of Failing Endpoints: 0

Total Number of Endpoints: 19

All user specified timing constraints are met.

VIVADO Flow



After implementation it is possible to check results in the Project Summary:

- Errors and warnings
- Resources Utilization (Graph or Table)
- Timing analysis
- Power consumption estimation

DRC Violations

Summary: 1 warning
[Implemented DRC Report](#)

Timing

[Setup](#) | [Hold](#) | [Pulse Width](#)

Worst Negative Slack (WNS): 3.121 ns
Total Negative Slack (TNS): 0 ns
Number of Failing Endpoints: 0
Total Number of Endpoints: 30
[Implemented Timing Report](#)

Utilization

[Post-Synthesis](#) | [Post-Implementation](#)

[Graph](#) | [Table](#)

Resource	Utilization	Available	Utilization %
LUT	129	17600	0.73
FF	30	35200	0.09
IO	16	100	16.00
BUFG	1	32	3.13

Power

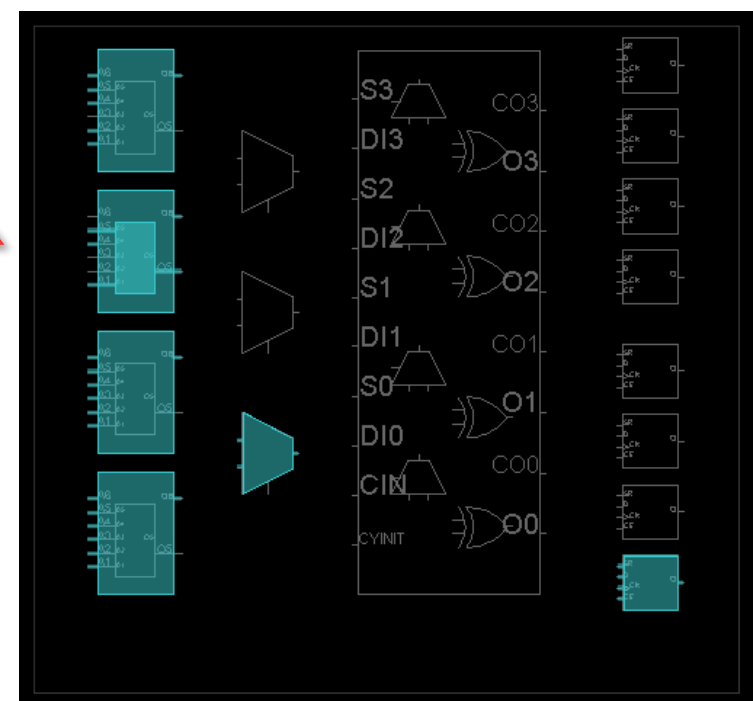
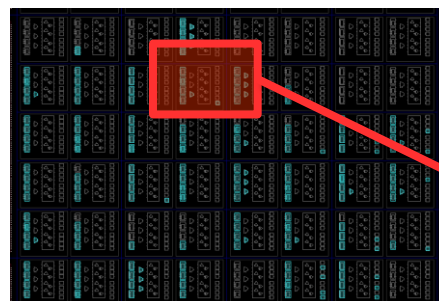
[Summary](#) | [On-Chip](#)

Total On-Chip Power: 0.109 W
Junction Temperature: 26,3 °C
Thermal Margin: 58,7 °C (5,0 W)
Effective θ_{JA} : 11,5 °C/W
Power supplied to off-chip devices: 0 W
Confidence level: [Low](#)
[Implemented Power Report](#)

VIVADO Flow



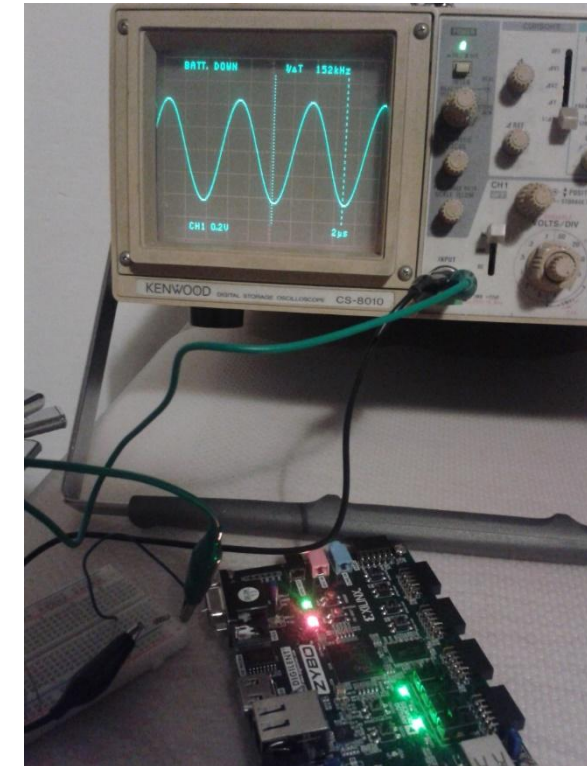
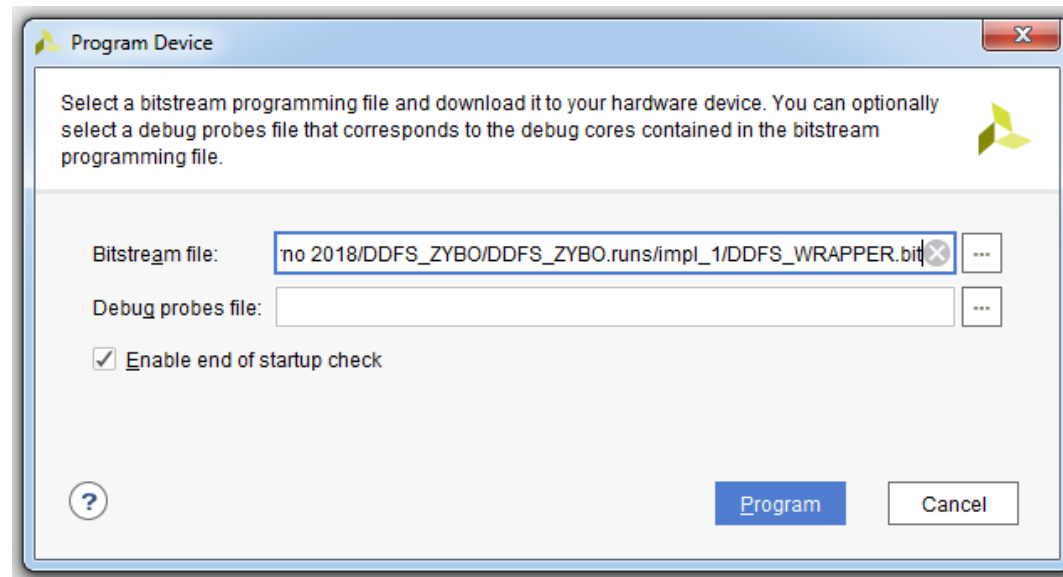
Open Implemented Design → «Device» window



VIVADO Flow



- Generate the bitstream file (*.bit)
- Connect the Device (ZyBo Board)
- Open Hardware Manager
- Program the Device
- Evaluate your DDFS



End, Questions?

- 1) ZyBo Development Board
- 2) Zynq 7000 APSoC
- 3) DDFS Implementation on ZyBo
- 4) Xilinx VIVADO Design Suite
- 5) Vivado Design Flow

